SUP'COM

المدرسة العليا للمواصلات بتونس
ÉCOLE SUPÉRIEURE DES COMMUNICATIONS DE TUNIS

Telecommunication engineering Degree
major:
Application of multimodal information (AIM)

Theme :

Cardiac abnormalities detection through Ballistocardiogram signals

---

Realized by
Mohamed Chiheb Ben Nasr

Supervised by
Prof. Sofia Ben Jebara
Prof. Neila Mezghani

Work proposed and realized within: CHUM reseach center

CRCHUM
CENTRE DE RECHERCHE

Abstract

This work enters with the framework of Telemedicine. This framework is rising in importance since it presents several advantages. One of which is the economic burden that presents the classical healthcare and especially overcoming the lack of specialized ICU beds [70].

This work concerns the detection of heart abnormality through the BallistoCardioGram signal (BCG). The latter signal is recorded with a Fiber Optical Sensor (FOS) placed under a smart connected mattress. To achieve the aforementioned goal of this work, different steps have been adopted during this work. The first step is the diagnosis and preprocessing of the available data set. The preprocessing concerned mainly three properties of the provided dataset: The noisiness, the skewness and the stationarity. Noisy frame detection as well as abnormal class augmentation has been conducted during this work. Studies concerning the stationarity of the provided signal were done and concluded with the existence of non statistically significant non stationarity in our signal. This property is particularly important since it rules over the choice of the forecasting model: subject of the first step of our modeling.

The first modeling step proposed a solution to forecast the signal frame several timestamps in the future. This has been achieved with the use of mathematical transformation as well as the ARMA model. The recorded errors found with the walkforward validation process was an sMAPE of 10%.

The second modeling step proposed is a solution to detect the cardiac events in a frame. This solution employs a Deep Learning (DL) model, mainly the Long Short Term Memory (LSTM) with Fully Connected (FC) layers. The LSTM plays the role of feature extractors and the automatically extracted features are fed to the FC layers for the binary classification task. This choice was done based on the intrinsic property of LSTM to keep the time dependencies. The results provided were expressed with respect to different metrics, some of which takes into consideration the unbalance of the provided dataset. An accuracy of 95% as well as a weighted accuracy of 96% was obtained.

The subject of future event detection through BCG signals is a novelty in the biomedical field and we hope this work will be a stepping stone towards future prospects and applications implementing the BCG signal.

Keywords— BCG, ARIMA model, LSTM, Time series, Connected mattress.

Acknowledgements

First of all, I want to thank everyone that helped me successfully achieve my graduation internship. Thank you for sticking with me throughout these hard times that the world lives in.

In particular I want to thank my supervisors for providing me with the guidance and help throughout these challenging months.

I want to thank Prof. Ben Jebara Sofia for her constant support and encouragement. In particular i want to thank her for her insights, her remarks and especially for pointing out my mistakes and telling me off when it had to be done.

I want to thank Prof. Mezghani Neila for providing me with this graduation internship and especially for giving me the possibility to contribute to a noble cause.

Advice given by Ouakrim Youssef has been a great assistance throughout this internship. I wish as well to acknowledge the help provided by the whole family of CRCHUM and LIO laboratory as well as the cheerful welcome and advice.

Special thanks to the 356 data science platform team for providing very educating and precious materials for free in the event of the Covid-19 pandemic. I also acknowledge the efforts provided by Jason Brownlee PhD in his blog entitled "machine learning mastery". Information provided in the latter blog are worth its weight in gold.

My warm thanks to the members of the jury for the honor they give me by accepting to evaluate my work.

Last but not least, i want to thank my family for the constant cheerleading, the moral and spiritual support and for being the backbone to every single success in my life.

Contents

Abstract **i**

Acknowledgements **iii**

1 General Context **2**

1.1 Motivation & objectives 2

1.2 Presentation of the hosting organisms 3

1.2.1 CRCHUM 3

1.2.2 LIO 4

1.2.3 COSIM 4

1.3 Problematic 4

1.3.1 The BCG signal 4

1.3.2 Acquisition and dataset 7

1.3.3 Target abnormalities 8

1.4 Related works 9

1.5 Description of the solution 13

1.5.1 Time series forecasting 14

1.5.2 Time series Classification 16

2 BCG signal preparation and checking for stationarity **18**

2.1 Data Preprocessing 18

2.1.1 BCG signal framing 19

2.1.2 Frame clustering 19

2.2 Data augmentation 20

2.3 BCG signal stationarity 23

2.3.1 Definition 23

2.3.2 Types of stationarity 23
2.3.3 Measuring the stationarity in BCG 24
2.4 Practical case 26
2.4.1 Seasonality example 26
2.4.2 Trend example 27
2.4.3 Stationarity results in the BCG signal 28
2.4.4 Stationarity results in the complete dataset 29

3 BCG signal modeling 31
3.1 The ARIMA model 31
3.2 Hyperparameter tuning 33
3.2.1 Tuning with autocorrelation function 33
3.2.2 Tuning with Likelyhood function 36
3.2.3 Residu analysis results 38
3.3 ARMA transformation using Box-Cox 39
3.4 Performance and evaluation 41
3.4.1 Error metrics 41
3.4.2 Walk-forward validation 43
3.4.3 Predictions and discussion 45

4 Event detection 48
4.1 Definition 48
4.2 Proposed solution 51
4.2.1 Model architecture 51
4.2.2 The magic within 52
4.3 Results 53
4.3.1 Classification metrics 53

5 Summary and Reflections 56

Bibliography 58

List of Tables

2.1 Cardinal of the classes before and after the data augmentation. 22
2.2 Obtained results of the different conducted tests. 28

4.1 Obtained metrics of weighted and augmented model and baseline model. . 54
4.2 Normalized confusion matrix of the obtained model. 54

List of Figures

1.1 Bradycardia obtained in Subject1. 5
1.2 Tachycardia obtained in Subject35. 6
1.3 Cardiac pause obtained in subject42. 6
1.4 ECG waveform segments [25]. 11
1.5 Proposed solution architecture. 14
1.6 sMAPE of different forecasting techniques [45]. 15

2.1 Preprocessing architecture. 20
2.2 Data augmentation architecture. 21
2.3 Spectral data augmentation. 22
2.4 Data partition for different abnormalities before the augmentation. 23
2.5 Evolution of the BCG over a period of 18s with p-values of 1.3211158767e-5 and 0.0723 for respectively the ADF and the KPSS tests. 27
2.6 Evolution of BCG over a period of 25s with p-values of 1.668241113e-5 and 0.0402 for respectively the ADF and the KPSS tests. 28
2.7 ADF values obtained from the different subjects(same critical values). . . . 29
2.8 p-values of the ADF test obtained from the different subjects). 29
2.9 p-values of the KPSS test obtained from the different subjects). 30
2.10 KPSS test obtained from the different subjects). 30

3.1 Autocorrelation function of the BCG signal 34
3.2 Partial autocorrelation function of the BCG signal 35
3.3 Evolution of the log likelihood over the number of parameters to fit 36
3.4 Evolution of the AIC over the number of parameters. 37
3.5 Acf of the Residual of the model. 38
3.6 Stationarity and its complexity. 39

3.7 QQ plot of residue *without* Cox-Box. 41
3.8 QQ plot of residue *with* Cox-Box. 41
3.9 Evolution of the sMAPE of the transformed ARMA model over the training size. 45
3.10 Prediction of 2 seconds ahead for the obtained model. 45

4.1 Basic LSTM cell 49
4.2 Adopted hybrid architecture. 51

Chapter 1

General Context

In this chapter we will provide the motivation and objectives of our work and a description of the Ballistocardiogram signal (BCG). We will also provide a general overview of some related works with the signal in question as well as some similar signals. In the last section we will present a general architecture of the solution we provided in during this internship.

1.1 Motivation & objectives

This work enters under the framework of telemedicine. Defined as the remote diagnosis and treatment of patients by means of telecommunication technology, telemedicine is gaining momentum these past few years and advancing with a steady pace towards becoming a standard in the medical domain [54]. This particular increase is mainly the effect of the advancements made in the IOT field. We forecast an increase of latter's importance with the probable release of 5G [5] [51].

The aforementioned intricate contexts motivates this work. This project concerns in particular a smart connected mattress and a Fiber Optical Sensor (FOS). The FOS is placed under the mattress, this sensor detects the movements of the body particularly the movements induced by the heart beats and the respiratory cycles [52]. The resulting signal is called the Ballistocardiogram (BCG) signal [23], subject of study in this work. The sought application of this study is the remote forecast of future heart abnormalities.

At the extent of our knowledge, few to no works have dealt with such problematic, this fact, added to the forecasted peak of interest in IOT applications in the future motivates this project.

We propose in this work an hybrid approach based on Statistical Modeling (SM) and Deep Learning (DL). The first part is employed to forecast the incoming BCG signal as it is shown that SM outperforms DL in this particular task [45]. DL was employed in the context of abnormality detection from the forecasted BCG frames. The goal from this adaptation is to be able to forecast lethal heart abnormalities that can occur to the patient whilst providing enough time for the medical intervention. In future works, we hope that with future works, we will be able to provide wider forecasting window of this detection.

1.2 Presentation of the hosting organisms

Three organisms have been involved in this work.

1.2.1 CRCHUM

The Research Center of the teaching Hospital of Montreal Hospital (CRCHUM) administers all of the research activities of the CHUM (University of Montreal Hospital Center). The research closely contributes to the hospital's five mission streams: care, teaching, research, health promotion and technology assessment. The CRCHUM contributes to the improvement of health in adults through a continuum of high-level university research which, by generating knowledge on the etiological and pathogenic mechanisms, promotes the development, implementation and evaluation of new strategies preventive, diagnostic and therapeutic. The CRCHUM offers a training environment ensuring a succession engaged in a search for excellence. More than 1,730 people work at CRCHUM, including 443 researchers and 420 graduate students. Each year, the researchers publish more than 750 scientific articles, or approximately 6 articles per regular researcher. This major production contributes to its excellent international reputation.

1.2.2 LIO

Today, health technologies call on multidisciplinary expertise. Those of the Orthopedic Image Laboratory include: morphofunctional assessment, regulatory requirements and quality management systems, material characterization, medical imaging and digital methods, geometric and bio-mechanical modeling, robotics, simulation and virtual reality.

1.2.3 COSIM

Communication, Signals and Image Research Laboratory (COSIM) was inaugurated in 2011 under the supervision of the ministry of higher education and scientific research. Containing approximately 60 researchers, COSIM is active in subjects concerning signal and image processing and numerical communication and its application on the national level. One of the main objectives of COSIM, which enters in the general context of this study is pathological signal processing. We mention as an example: voice signals, cardiac signals such as ElectroCardioGram (ECG) signals, movement signals as ElectoMyoGram (EMG) signals and mechanical signals. The aforementioned signals are studied in order to understand the behaviour of the human body, facilitate the medical diagnosis and boosting the athlete's performance.

1.3 Problematic

1.3.1 The BCG signal

The BCG signal captures the results of the ejection of the blood during the systole. This result generally manifests through the mechanical reaction of the body. In particular, the movement of the thoracic cage for the Respiratory Rate (RR) as well as the Heart Beat Rate (HBR) are captured using a Fiber Optical Sensor (FOS) and the result is the signal at hand. Several studies have been conducted in the subject of signal processing and machine learning over the BCG signal. We mention [10] where the latter is used to detect several respiratory activities such as normal respiration or coughing and we can

also refer to [52] for the extraction of the HBR as well as the RR from BCG signals where different signal processing algorithms has been utilized to extract the latter rhythms.

However, the problem with BCG signal, has and will always be the absence of a unified and worldwide waveform as opposed to the PQRST complex in ECG signal.

Another particular barrier in the face of the widespread of the BCG signal is its non invasive nature. While such nature is highly appreciated in the field of medicine since it provides a certain liberty when it comes to the implementation of the device, it induces lots of movement originated artefacts that prove to be difficult to handle.
Several works have been conducted in this subject, we mention [7] where a trial and error based methodology have been implemented to denoise the BCG signal using Multi-resolution analysis in particular wavelet transforms as well as corresponding ECG signals as reference. Other works implement advanced methods such as in [12] where an independent component analysis approach has been implemented in order to detect movement artifacts.

Many other works have been done in this particular subject such as the detection of useful frames using spectral descriptors namely Spectral Flatness Measure (SFM) and Spectral Centroid (SC) [9].

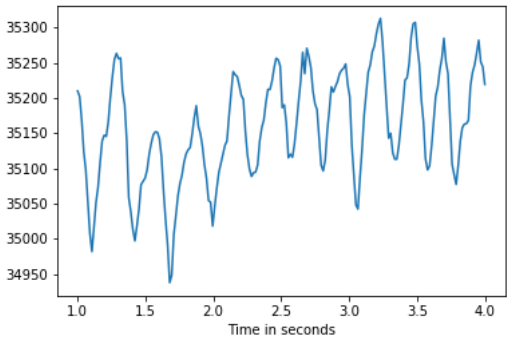


Figure 1.1: Bradycardia obtained in Subject1.

Figure 1.1 shows a case of a heart abnormality, subject of this work. In this figure, the reference ECG signal recorded a HBR of 37BPM. This is considered a low cardiac rhythm and thus one of the studied abnormalities. This low HBR can be extracted from the BCG signal using different techniques (this was the subject of a past study done

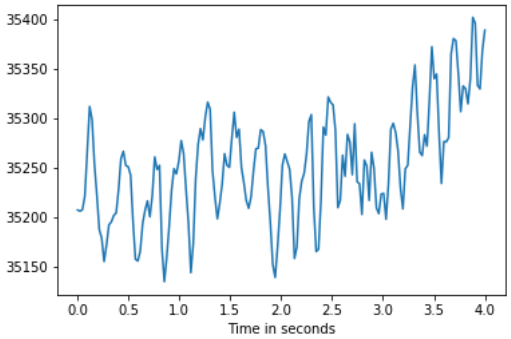


Figure 1.2: Tachycardia obtained in Subject35.

in the event of the sophomore internship [8]), mainly extracting the little fluctuations that exists within the respiratory activity. Figure 1.2 shows a case of another type of abnormality which is tachycardia. An elevated HBR was recorded 168HBR which is a sign of arrhythmia. We can refer to the time interval [2.5s, 3s] for such an abnormality. The rapid fluctuation of the heart captured by the FOS results in this particular portion. Figure 1.3 showcases a brief cardiac arrest recorded in the 42nd subject. The heart of this subject stopped for approximately 1.5s. This can be observed in the aforementioned figure when referring to the interval [1s, 3s]. We want from our sought solution to take into consideration these particular variations in the BCG signal at a first level and correctly annotate them in future works.

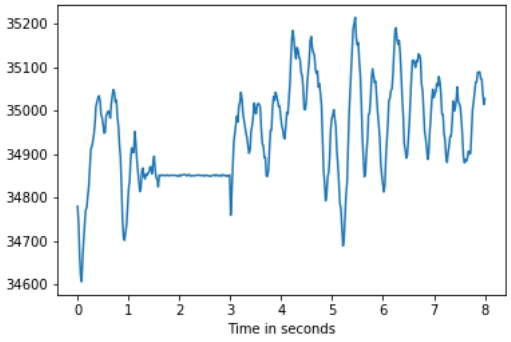


Figure 1.3: Cardiac pause obtained in subject42.

1.3.2 Acquisition and dataset

The main aim of this work is the detection of several cardiac abnormalities using the provided BCG signal.

In fact, recorded from 100 subjects, we are presented with a dataset of input signals that describes the cardiac and respiratory activity of the corresponding patients during a period of 24 hours. The recorded signals are made available to us with a FOS placed under the mattress of these patients during their stay at the CHUM.
In the meanwhile, a device that generates paper-based reports of the cardiac abnormalities occurring during the stay of the patient is wired and takes the ECG signal of the patient as a reference. The reports are then digitized using an Optical Characters Recognition (OCR) algorithm and temporally synchronized using the Unix timestamp convention.

The use of OCR, however induces multiple mistakes in the generated data, mistakes that originate from some similarities between some characters, lighting, writing style... These mistakes had to be manually filtered and taken care of in order to rigorously detect the abnormalities later on.

Another property to highlight in our dataset is its skewness. In fact, taking into consideration only the normal frames of the first subject and the abnormalities from all the 100 subjects, we find ourselves with a highly unbalanced data with an unbalance ratio of 43000 to 120 which translates to: we are presented with 43000 frames of normal frames and only 120 frames in which a cardiac abnormality manifests. We need to keep in mind that these proportions would blow up if we take into account the normal frames from the 100 subjects similarly to what we have done with the abnormal frames (a ratio of 4000000 to 120).

This particular fact is very understandable and reasonable and reflects the physical reality. For example, during the stay of a subject in the hospital cross-referenced with the available recording input, the frame length and the sampling we do not expect more than 4 cardiac abnormality to occur per patient. Some severe cases that exceed this quota happen to be very critical cases and we do not have the luxury to record their BCG signal. The target cardiac abnormalities of this work will be further discussed in the next

section.

1.3.3 Target abnormalities

In this work we deal with six different abnormalities which are listed and described as follows.

Tachycardia

It is one of the most common types of cardiac abnormalities. This particular abnormality refers to a very high HBR of 90 to 160 BPM during rest. Note that high HBR is perfectly normal for a person to experience such as during physical activities, stress, trauma. There are different types of tachycardia[2], we relate for example atrial, benign, ventricular, bidirectional ventricular, ectopic, junctional,...[3]

Bradycardia

Similarly to Tachycardia, Bradycardia is another type of arrhythmia that describes a critically low HBR. In fact, a HBR below 60 BMP is considered to be critically low and a symptom of Bradycardia. Bradycardia is a very serious condition since the heart does not pump enough oxygen-rich blood to the rest of the organs. This can result in the malfunction of the organs and can be corrected using a pacemaker[1].
Both Tachycardia as well as Bradycardia can be detected through the measure of the HBR which is possible as mentioned in the previous section through the BCG signal.

One of the variants of the bradycardia that we are studying in this work is ventricular bradycardia. This is due to the lack of electrical impulses from the atrium.

Cardiac pause

Also referred to as sinoatrial arrest. It corresponds to a condition where the heart ceases from generating electrical impulses and thus the heart ceases from beating. This effect can last up to several minutes. This condition can be observed from the BCG signal by the absence of the cardiac activity fluctuations.

Atrial fibrillation

It is very similar to tachycardia, the HBR is irregular and very often high. This particular condition can result in blood clots in the upper chambers of the heart. This will result in the blocking of the blood flow in the organs which will result the long term, in a ischemia [30].

Asystole

It is another type of arrhythmia. However, this abnormality can be the result of the lack of ventricular contractions. This is the most lethal type of cardiac arrest and can result in irreversible damage to the heart tissues. Asystole is very similar to the cardiac pause, the consequences are, however, dramatically different. The main cause is a prolonged sinus arrest.

For each of these abnormalities we are presented with the corresponding BCG variation recorded. The aim of this work is to be able, in the first place, to forecast the presence of a heart abnormality and set up the environment for the detection of the specific heart abnormality in the long term. The input of our setup will take the form of a time series and the output is the presence/absence of a heart abnormality.

1.4 Related works

In this section, we will be reviewing some of the existent works in the field of forecasting biomedical time series. An extended review showed that a very preponderant portion of the works in the field has been done on electrocardiogram signals (ECG).
This focus on the ECG signals is motivated by different reasons, the biggest of which is the global use of these signals as the main medical tool for the diagnosis of heart abnormalities.

At the extent of our knowledge, few to no works has been done on BCG signals forecasting which makes the subject of this internship very attractive since it can be labeled as a novelty. We hope that this work will smooth the path towards more attention

to the BCG signals since, with the rise of connected objects and 5G technologies in the next few years, we anticipate a keen interest to non invasive connected setups, in particular BCG sensors.

The huge similarity between BCG signals and ECG signals, as well as the conformity and the developed research conducted on ECG signals motivated the use of such works as reference for this project. These resemblances between the two counterparts are studies in multiple works for example in [49] 95% of BCG based extractions (R wave, QS wave, zero level and PT wave) were found within the 1.95 standard deviation range of its ECG counterparts.

In the context of ECG forecasting, we cite [39] where the authors are predicting the internal heart rate: an intrinsic characteristic of the fetus's heart activity, using a multivariate auto regressive model. The different variables are extracted from the ECG signal using wavelet transformation. The interesting part about this work is using the same ECG signal to extract different variables that contain different information about the sought target. We can in fact see the improvement of the accuracy by the employment of the wavelet based techniques instead of the row ECG signal. By extracting multiple signals from the ECG, and deploying the model by the use of these signals instead of the original ECG, a very remarkable improvement in performance was recorded, especially when it comes to multiple steps forecasting.

Other works are [61] where a mix of Back Propagation Neural Network (BPNN) and Variational Model Decomposition (VMD) was used. In fact BPNN can predict non linear and non stationary signals, they, however, present several drawbacks one of which is the hyper parameters tuning (see section 3.2). VMD was used to act on this particular issue as well as other improvements such as noise cancelling, influence of subjective factors and input transformation. The influence of this technique has been recorded using different indexes such as Signal to Noise Ratio (SNR), Root Mean Squared Error (RMSE) and correlation coefficient (COR).

In [20], the authors propose an hybrid model. A mix of cross sectional models and time series models are implemented to predict the wellness of the subjects for up to one day

ahead. ECG signals go through a preprossessing phase where noise filtering, segmentation and normalization are performed. A 5 second long segment is fed to 2 Deep learning models: Long Short Term Memory (LSTM) and Bidirectional LSTM models (BiLSTM). The output is coupled with the results from an Artificial Neural Network (ANN) as well as a Support Vector Machine (SVM): two machine learning based models fed with input data of Health Indexes (HI) of the patients. While Machine learning based models shows an accuracy of 62%, deep learning based methods show very promising results reaching an accuracy of 90%. These promising results motivate the use of time series in medical applications instead of cross sectional data.

Other works drift from the concept of utilizing time series such as those of [57] which makes use of Dynamic Time Wrapping (DTW) for the purpose of denoising the signal as well as for feature extraction. These features being the amplitudes and locations of each of the ECG signal Waves (R wave, QS wave, zero level and PT wave).

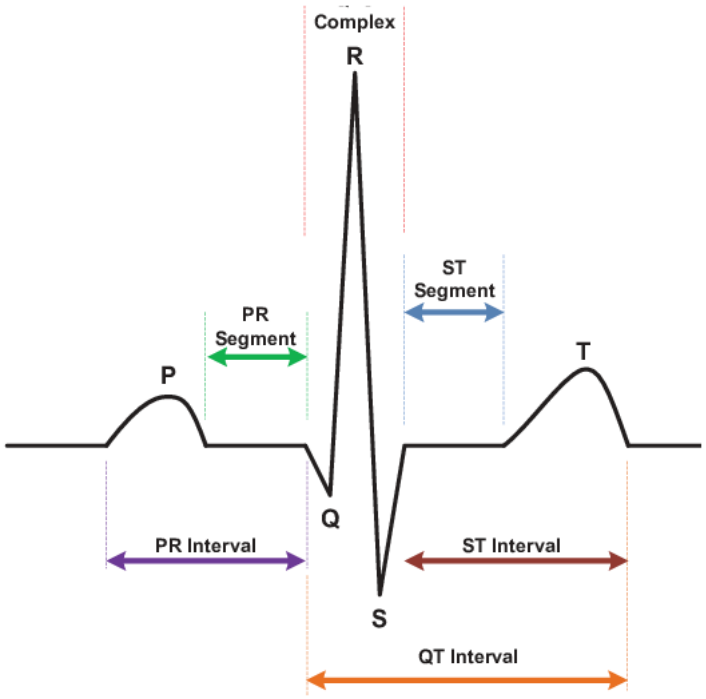


Figure 1.4: ECG waveform segments [25].

The work in [37] takes into consideration a very promising and novel aspect in machine learning which is transfer learning (TL). TL is the implementation of previous knowledge of a certain domain for specific application. In the example of Convolutional Neural

Networks (CNN), the first weights of the Convolutional Layers (CL) contain general information about the subjects that will be conducted in practically the same fashion for every classification application. The first weights contain, for instance, information about the extraction of the basic patterns of the signal. Such information is very crucial for the classification. The particular interest in [37] is to create transferable deep learning representation of the ECG signals since there is a crucial lack when it comes to transfer learning for time series. The architecture implemented is a mix of CNN and Fully Connected (FC) layers, where the first layers of the CNN part can be reused for very similar purposes. In fact, the visualization of the output of these layers shows a very separable representation which strongly favors the discrimination between the classes of the proposed Arrhythmia.

In the subject of pattern recognition, some works have been conducted on BCG signals. We particularly mention [44] where the purpose is the classification of Arrhythmia using Cumulative Residual Entropy (CRE) as a feature extraction tool and SVM as an inference tool.

Other works such as that of [43] implements deep learning methods combined with Extreme Learning Machine (ELM)[4] for the detection of the IJK complex in BCG signals. The CNN architecture is based on CL as well as Max pooling layers for the feature extraction process. This architecture shows very promising results especially for the HBR detection from the BCG signals. In fact, consecutive BCG segments are fed to the CNN for the detection of the IJK complex and the information of interest is the time measured between two positive detections. Hence, the HBR can be measured.

An extended work in [22] showed very promising results when it comes to the detection of Obstructive Sleep Apnea (OSA). The feature extraction phase has been done manually to include frequency and time domain features such as statistical descriptors (mean, variance, standard deviation of NN intervals) or spectral power based descriptors (mainly power recorded in different bands). Decision tree based models as well as linear regression model and SVM have been implemented and combined to boost the performance of the classifier. a K-fold cross validation process has been performed to obtain a final accuracy of 89%.

Finally, in [55] a 1d-CNN combined with BiLSTM has been implemented to detect with an accuracy of 74%, the different sleep stages. The performance of the model is measured in this application by the use of the correlation coefficient with the SATED sleep quality (Satisfaction, Alertness, Timing, Efficiency and Duration) scores with a value of 0.43, a little shy of the baseline performance recorded for the PSG method: the gold standard when it comes to diagnosing sleep-related problems.
In the next section we will discuss the solution that we propose for our problem as well as the motivation behind the architectures presented.

1.5 Description of the solution

In this section, we will describe the solution that we are going to implement.
The first step is to decompose the problem into two different parts. In fact, we are asked to predict, using a BCG time series, a possible future cardiac abnormality in the patient. Future being a keyword is this work. We are not asked to identify the present abnormalities such as in the previously mentioned works dealing with classification but to forecast such problems in the future in order to establish the sufficient procedures to prevent the deterioration of the patient's health.
This point is particularly important in the framework of connected objects since knowing N timestamps ahead of the possibility of occurring of an abnormality might give enough time for the medical stuff to intervene.
The second part of our solution is the pattern recognition part. Given a certain time series how can we detect the presence of a certain abnormality in it.

These two steps are resumed in figure 1.5, the input is the time series from instant $t-m$ to instant t : $\{x_{t-m}..x_t\}$. The forecasting will deliver the future previsions of $\{\hat{x_{t-m+s}}..\hat{x_{t+s}}\}$ where s constitutes the future lag.
Next, from the forecasting model output, a classifier will be used to the coming event. For instance, a coarse classifier will detect if the future event will be normal or abnormal. A finer classifier will give the kind of abnormality if it exists (tachycardia, bradycardia, ...). These two steps will be detailed in the following subsection.

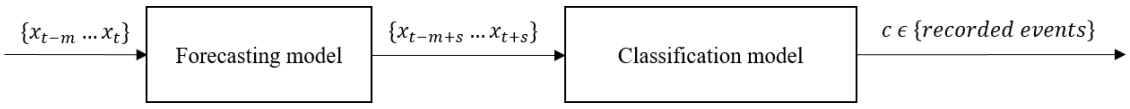


Figure 1.5: Proposed solution architecture.

1.5.1 Time series forecasting

This is the first part of the proposed solution. A device placed under a connected mattress of a patient lying at home will provide us, in a continuous manner, with the BCG signal of the subject. After framing it, the first part of the proposed solution will begin to act. Provided with a time series of length $m = 200$: $X_{t-m}...X_t$ the first model will forecast our signal $s > 0$ refers to the future lags which means it will provide predictions of the value of $X_{t-m+s}...X_{t+s}$. This particular task is referred to in the literature as a forecasting step. Different techniques treat such problem, we relate in particular Deep Learning (DL) and Statistical Modeling (SM). Even though DL is rising in popularity in the last few years and provides very complicated and interesting solutions [64] [28], its results are still shy of those recorded using SM algorithms.

In this study, we chose to use a modified version of the Auto Regressive Moving Average (ARIMA) model which is the transformed ARMA model. This choice has been made since the latter model outperforms novel deep learning methods when it comes to forecasting time series methods [45] [29].

As illustration of different forecasting techniques' performances, Figure 1.6 shows the evolution of sMAPE in a variety of applications and from different research topics. sMAPE is standing for symmetric Mean Absolute Percentage Error which is the most commonly used error metric in forecasting. Figure 1.6 shows that The ARIMA models outperform the majority of the other models such as the LSTM models as well as machine learning approaches. This fact is explained by the high variance problem that commonly occurs in the LSTM models as well as the fact that deep learning techniques are more computationally expensive [45], more data hungry and prone to overfitting [56]. However, this does not reflect our personal take on these models, since ARIMA models are only able to model linear relationships in the time series, while the LSTM can model more complex

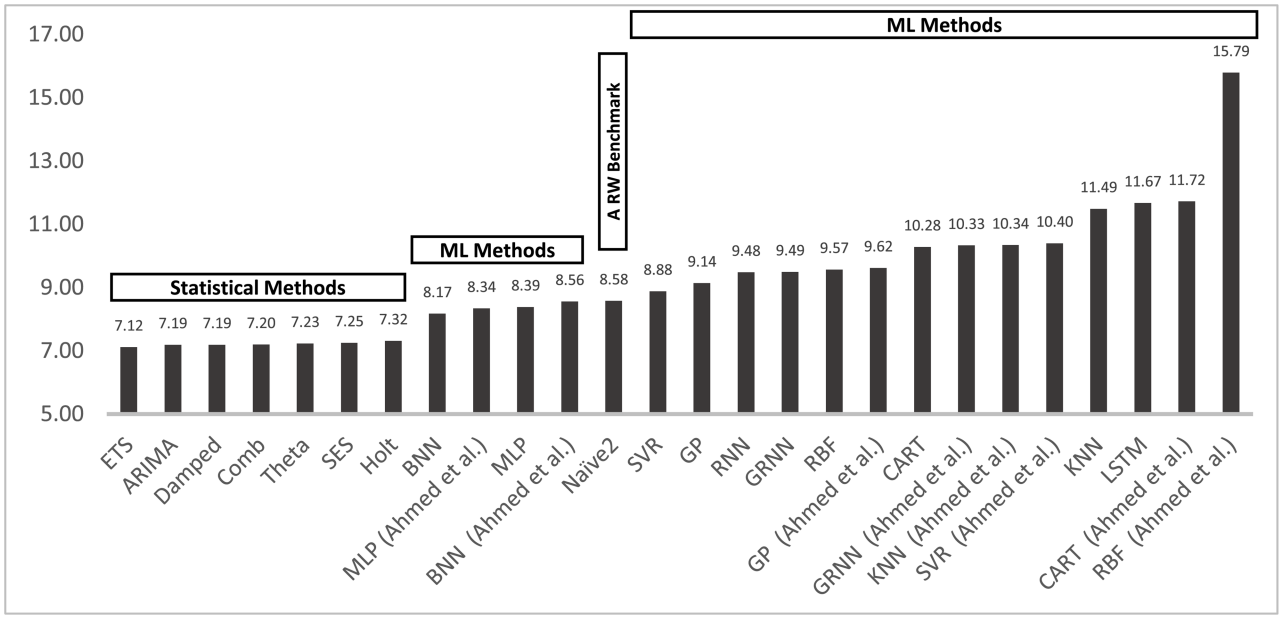


Figure 1.6: sMAPE of different forecasting techniques [45].

relationships at the expense of the computation complexity. This choice has been made since the procedure of forecasting in the BCG signal is a novelty and at the extent of our knowledge, few works deals with such a use case. So for the sake of the scientific intuition we chose to start with a less complex and more physically interpretative model to fit our data in.

Another point that motivated this choice is the fact that we are able to provide a mathematical formulation of the hyperparameters tuning problem when it comes to ARIMA models as opposed to the trial and error based techniques in deep learning. We provided detailed explanation the section 3.2.

In [45] which is one of the most known forecasting references, different forecasting methods has been tried. They were given the same preprocessing and evaluated in terms of generalization errors, goodness of fit and computational complexity. This further boosts our confidence to achieve very respected results using linear models. A more detailed description will be provided in section 3.1 for this modeling step.

1.5.2 Time series Classification

In this subsection, we will discuss the second part of the proposed modeling. This part concerns the pattern recognition step of the model. In fact, given a time series $X_t...X_{t+200}$, we seek to classify the BCG data to normal/abnormal classes and eventually to identify different types of abnormalities.

In the literature, multiple approaches can be utilized to achieve this goal of time series classification. For example, signal processing techniques such as Dynamic Time Wrapping (DWT) [34] . The latter can provide an advanced temporal description of similarities between time series without taking in consideration their paces. This can provide, compared with reference values of DWT of different abnormalities decent results.

Other machine learning based methods can be employed. We can, similarly to the work done in [22] manually extract temporal and spectral features and feed them to a common classifier such as SVM that will infer the corresponding class. However, the temporal aspect of the signal motivates the use of DL techniques that captures the time dependency rather than cross sectional models. The most commonly used models are Recurrent Neural Network (RNN) and its variants.

Transfer Learning (TL) is always a possibility but due to lack of models featuring biomedical time series we judged it unnecessary and rush especially giving the novel aspect of our work.

Some works implement an hybrid architecture between RNN and CNN and use attention mechanisms [38]. These models however tend to be very complicated and are hence deprecated as first implementations. In this work, we chose to implement an LSTM architecture coupled by a Fully Connected (FC) layer. the LSTM layers are followed by dropout layers to avoid overfitting. These layers will randomly drop a percentage of their weights in order to avoid the high variance issue. The choice of LSTM over the other RNN architectures is due to the vanishing gradient problem. This problem will be further discussed in chapter 4.

To resume, classic RNN architectures are not able to model information further in the past the rule of thumb is a maximum of 3 timestamps in the past [31] due to a

stacking of multiplication operation of Sigmoid functions where the output is less than 1. Thus, no matter the value of the output it will not stand the test of time (pun intended). And finally different approaches has been conducted for the hyperparameters optimisation that will be discussed further in subsection 3.2. Errors and classification reports will be presented in the section 4.

Chapter 2

BCG signal preparation and checking for stationarity

This section will concern the different types of transformations executed on the available BCG signal. In particular, we will provide the different data preprocessing techniques used in this work, as well as a section treating a very important issue in the subject of time series analysis which is stationarity. This concept is very interesting, since depending on the results, we can choose to exclude a certain component of the ARIMA model and thus further simplify its formulation.

2.1 Data Preprocessing

One of the most important steps in the pipeline of a data science project is data preparation. This step consists on transforming and verifying the data in such a way that it yields the best possible insight of the sought application. In this work, three big properties of our row signal emerge: the skewness data-wise, the noisiness and the stationarity. Therefore, this chapter deals with these aspects. The preprocessing phase was focused on the selection of the frames that doesn't contain movements of the subject and thus yield significant information about the cardiac and respiratory activities [9] as well as data augmentation that aims to tackle the skewness of our data and its relatively small size regarding abnormal activities (see section 2.2).

2.1.1 BCG signal framing

The BCG signal is sampled at the rate of 50 Hz and we settled at 200 as the length frame. This choice has been done according to a brief literature review of the minimal time needed to detect the different cardiac abnormalities, mainly the Bradycardia since we need a very pertinent lower bound of the frame size to be able to confidently conclude on pathology. For example, in the case of Bradycardia, for which the HBR is low resulting in longer cardiac cycle, the choice of a very low length frame can lead to short frames not containing a whole cardiac cycle. For an HBR of 60 BPM, the cardiac cycle will take approximately 1s and thus 50 samples. If we choose 4 cardiac cycle as the baseline that gives statistically significant results a frame size of 200 samples is quite sufficient.
Another factor to take into consideration is the overlap of the classes. Apart from the statistical significance translated in at least 4 cardiac cycles, the upper bound of our time frames is ruled by the overlapping of classes. We do not want to implement a frame lengthy enough to contain multiple classes since we do not know how to label them. For example, a frame size that allows the segment to contains multiple abnormalities is judged to be a bad choice since in that case, we are unable to correctly annotate the frame. The work done during the sophomore project treats in detail this particular issue and proposes a studied range of frame sizes that obeys to such hypothesis [9].

2.1.2 Frame clustering

Different techniques has been adopted to denoise the BCG signals [8]. The implementation of such techniques are mainly implemented through filtering. Figure 2.1 shows the architecture of the adopted solution. A BCG frame of length 200 samples (equivalent to 4 seconds of recording) is the input of a feature extraction phase, where both the Spectrum Flatness Measure SFM and the Spectral Centroid (SC) are extracted from the corresponding frame. The extracted values are then fed to a trained GMM where the binary output corresponds to the adoption/discarding of the frames. Using this filtering approach, we can be provided with the information of the corruption of the frame and

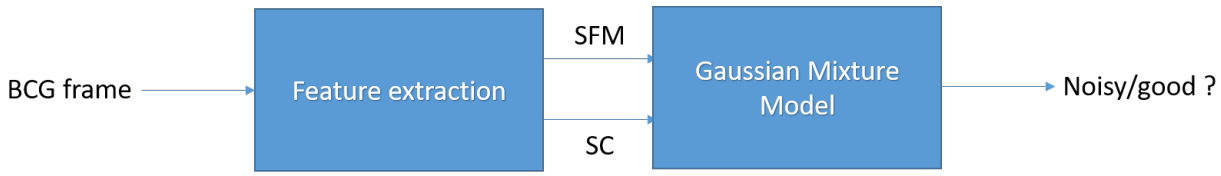


Figure 2.1: Preprocessing architecture.

thus we can chose to discard the corresponding frame or maintain it with certain credibility scores. This work has been done during the sophomore internship [9] . Using features as SFM and SC as well as an unsupervised learning algorithm namely Gaussian Mixture Model (GMM) we were able to distinguish between noisy and unnoisy frames with a high accuracy [10]. Such method can be adopted in order to single out frames where the significant cardiac information is perturbed with a certain artefact, namely the patient's movement. These features are based on the periodicity of the input signal and can hence detect the existence and the consistency of the IJK complex in the input frame. Other filtering techniques can be certainly adopted for noise filtering, works such as in [42] or [58] use Kalman filters to detect and remove the movement artefacts induced in different signal types using acceleration signals. This particular signal is made available to us in this study, however, fault of time restrictions and project complexity we were not able to implement such advanced techniques. Another possibility is to implement a median filter as a denoising tool, this way we can get rid of outiers in our signal. This technique is mostly notorious in 2D domains such as image processing and often referred to as salt and pepper noise [72]. Some works implement such filters in the 1D domain, namely for time series denoising [59] where such technique has been implemented for chemistry applications.

2.2 Data augmentation

As mentioned in the previous sections, during this work we implemented a low-level solution to overcome the small dataset size and the skewness of data distribution between classes. This solution is referred to as data augmentation in the literature. The main approach we took in this project is the oversampling. It consists of adding a certain

transformation to the available data in order to augment the cardinal of the dataset while keeping the same defining characteristics. The main data augmentation approach used in this work is adding white noise. This approach grants us a certain augmentation of the cardinal of our data as well as particular robustness to the white noise that can exist in natural conditions through the heating of the materials as well as other perturbations that can be modeled as white noise. This double utility of the utilization of white noise is very attractive in our work since we are presented with a skewness ratio of 43000 to 120. In fact, the considered dataset contains 43000 frames of normal activity and only 120 frames of abnormal activity. There's, however, a certain limitation to this technique since mindlessly adding noise will in certain cases add a bias to our model. In this work, we studied a literature review on data augmentation in time series analysis [53] and we chose to implement basic approaches, namely in temporal and frequency domains. This combination has been proved to yield significant improvements [21] especially in the anomaly detection problems. In the figure 2.2, a description of the data augmentation

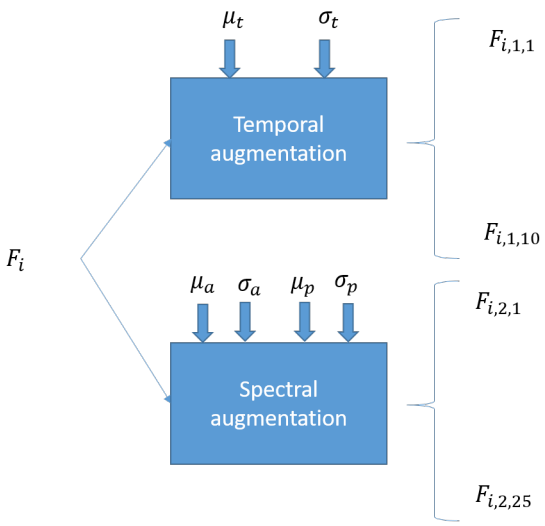


Figure 2.2: Data augmentation architecture.

process is provided. The frame labeled F_i of length 200 is introduced to a temporal and spectral augmentation processes. In the time domain, a white noise has been introduced with mean μ_t and variance σ_t to output 10 different temporal signals. In the frequency domain, a white noise has been introduced into the amplitude as well as the phase of the spectrum, with an amplitude's mean of μ_a and variance σ_a as well as a phase's mean of

Table 2.1: Cardinal of the classes before and after the data augmentation.

	Before augmentation	After augmentation
No abnormality	43000	43000
Cardiac abnormality	120	4200

$\mu_p = 0$ and variance σ_p. The values of μ_t, μ_a, σ_t, σ_a and σ_p has been chosen manually with respect to the mean and variance of the signal. The chosen values are under the limit of 0.1 of its original counterparts. Figure 2.3 shows the procedure of adding white

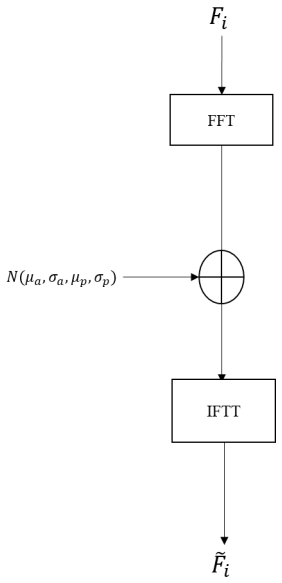


Figure 2.3: Spectral data augmentation.

noise in the frequency domain.

In table 2.1 we provide a detailed partition of the before and after augmentation. For each frame belonging to abnormality class, 35 noisy frames are generated. 10 of which are generated through temporal augmentation and the rest is obtained through spectral augmentation. However, even after data augmentation, the data is still skewed: a ratio of 1 to 10 is obtained this time. Another solution will be adopted in the section 4. It is worth mentioning that in the framework of future works, where we are seeking to migrate our solution from binary classification where the task is to simply predict the existence of an abnormality, to a multi-class classification where we are asked to pinpoint the particular cardiac irregularity, the distribution of the abnormalities is not equal and we can see a high skewness towards the Bradycardia class. This is particularly shown in figure 2.4. The latter might trigger an interest to augment some classes at the expense of others or

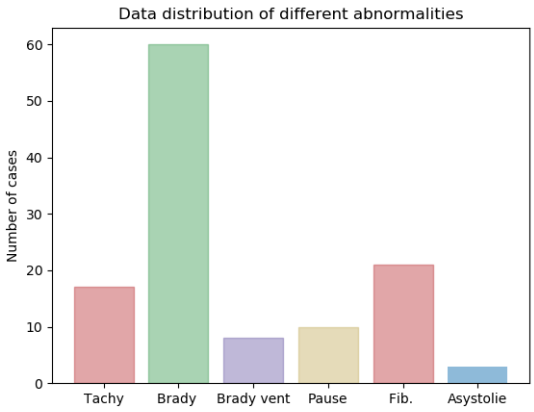


Figure 2.4: Data partition for different abnormalities before the augmentation.

to treat the problem in a different manner (anomaly detection).

2.3 BCG signal stationarity

2.3.1 Definition

One of the most important concepts in time series analysis is the concept of stationarity. An informal definition of stationarity is that the statistical properties of a process generating a time series does not change over time. This is extremely useful when it comes to statistical modelling since it denies the existence of an external cause that rules over the evolution of the time series (trend and seasonality [60]). The formal definition is divided into two types of stationarity, A formal definition of this concept as well as practical results will be provided in the following sections.

2.3.2 Types of stationarity

Strong/strict stationarity

Defined as the shift-invariance in time of the finite dimensional distribution of a stochastic process $\{X = x_i(w), i \in Z\}$ where x_i is a family of random variable defined on the same probability space (Ω,F,P).

$$\{F_X(x_{t_{1+\tau}}, .., x_{t_{n+\tau}}) = F_X(x_{t_1}, .., x_{t_n}); n \in N^*, \tau \in Z\}. \tag{2.1}$$

Where F_X is the finite dimensional distribution defined as:

$$F_{t_1,..t_n}(x_{t_1}, .., x_{t_n}) = P(X_{t_1}(w) < x_{t_1}, .., X_{t_n}(w) < x_{t_n}). \tag{2.2}$$

This, however, is a very strong hypothesis to test especially when it comes to computational power and condition of realisation in the nature [15] and is in most case not valid. More details and discussion was made in chapter 3. This points us at the weak stationarity definition.

Weak stationarity

This property requires only the time-invariance of the mean and the autocorrelation of the process which depends only on the difference of time between the points as well as a finite variance:

$$R_{X,X}(t_u, t_v) = R_{X,X}(t_{u+a}, t_{v+a}). \tag{2.3}$$

Where $u, v \in N, a \in Z$.

Trend stationarity

A process is trend-stationary if an underlying trend (function of only time) can be removed, leaving a stationary process. Meaning that the process can be expressed as :

$$y_i = f(i) + \epsilon_i. \tag{2.4}$$

Where f is a function from R to R and ϵ_i is a stationary process with zero mean.

2.3.3 Measuring the stationarity in BCG

Although the task of stationarity detection seems a straightforward task with the available equations, this is not applicable in real life since checking the stationarity through its

formal definition is a computationally expensive task (see section 3). That's why the procedures employed in time series analysis use statistical tests. Two different tests were deployed during this work: the Augmented Dickey Fuller test (ADF) and the Kwiatkowski-Phillips-Schmidt-Shin test (KPSS).

The Augmented Dickey Fuller Test (ADF)

The simple Dickey Fuller test checks for the unit root. It means that this test checks whether our process is a random walk which is the null hypothesis of the test. In the case of the confirmation of the null hypothesis, the time series is a random walk and thus not stationary. The introduction of the ADF [47] generalizes the former hypothesis to the case of multiple time steps behind as opposed to the random walk that only includes the previous lag. In fact, the random-walk of a time series x_t can be expressed as follows:

$$X_t = c + \phi X_{t-1} + W_t. \tag{2.5}$$

Where ϕ is the pondering coefficient and W_t is the additive white noise. The Markovian model at question in this test is the following

$$X_t = c + \sum_{i=1}^{n} \rho_i X_{t-i} + W_t. \tag{2.6}$$

Where c is a constant, X_t is the time series and ρ_i are the coefficients that the we will test. The ADF tests whether the coefficients ρ_i are close to the unit. In the case of n=1 we can see the particular case of a random walk. We can check the results of the test by examining the p-value of the null hypothesis. The threshold of p-value used in the medical field is 0.05. Another method to use is to compare the test value with the critical values provided at different thresholds: 1%, 5% and 10%. Note that the least values of the critical thresholds imply a more restrictive conditions i.e. if the test value is less than the critical value at 1% then it satisfies to the condition of 5% as well as 10%. If we want to further explain the physical significance of the critical values, we can say that a critical value of 10% means that we are only 10% sure that the null hypothesis is due to a certain

phenomenon in the signal and not induced from noise and thus we are 10% sure that our signal is not stationary. This particular point is worth criticizing (it has always been a subject of discussing and division between statisticians).

Kwiatkowski-Phillips-Schmidt-Shin test (KPSS)

The main use of this test is to check for trend stationarity as well as random walks (not generalized) in the case of $\rho = 1$ which is a very important aspect in time series.

$$X_t = c + \rho X_{t-1} + \beta t + W_t. \tag{2.7}$$

Unlike the Augmented Dickey Fuller test, the null hypothesis of this test is that our signal is trend stationary which implicates the non-stationarity. This particular type of stationarity, however, can be handled using different kernels in the trend option present in the majority of the models (models can handle the trend stationarity by modeling it by testing different hypothesis). The test of the hypothesis is confirmed by obtaining p-values greater than 0.05.

2.4 Practical case

In this section, we chose to display different stationarity examples in our data set. Two major types are studied here which are the seasonality and trend.

2.4.1 Seasonality example

In our dealing with BCG signal, stationarity tests are carried. After performing these two tests, we can confirm through the p-values that our signal satisfies the ADF as well as the KPSS tests. However these results do not imply the complete stationarity of the signal we can for example mention the presence of cycles in our time series. These cycles are present due to the cardiac and respiratory activities of the subject which is an important information in our analysis. These cycles are however different from the common type of periodicity that provides a certain level of non stationarity, often referred

to in the literature as seasonality [60]. Figure 2.5 shows that our signal presents a certain

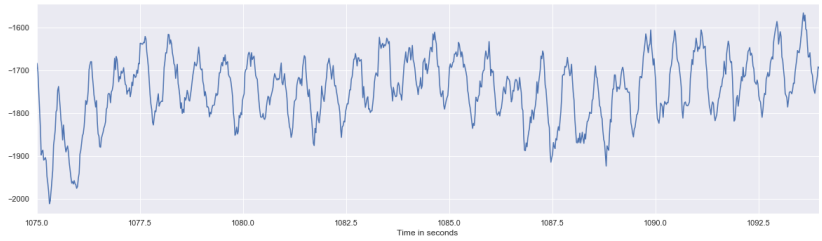


Figure 2.5: Evolution of the BCG over a period of 18s with p-values of 1.3211158767e-5 and 0.0723 for respectively the ADF and the KPSS tests.

seasonality-like pattern. Seasonality [60] is a technical term used in statistics to identify periodic behavior of the signal over time. The presence of a seasonality as well as the trend implies the presence of non-stationary behavior in the signal. This type of non-stationarity is not accounted for in the previously mentioned tests. We can observe the carrier-like behavior of our signal induces peaks around the 1077s, 1084s, 1091s as well as troughs around the 1075s, 1081s, 1088s. The rule of thumb when it comes to such a behavior is to manually tune the models to account for such cyclic behavior. A general remark in this section is concerning the source of this behavior. It is believed that the high frequency fluctuations are the result of the cardiac activity of the subject while the low frequency variations originate from the respiratory activity of the subject. If we want to interpret the results, the obtained p-values of both the ADF as well as the KPSS test implies the stationarity of our signal, by rejecting the null hypothesis in the first one and admitting it in the second.

2.4.2 Trend example

We chose to include Figure 2.6 to showcase some other types of non-stationary behavior in our signal. We can observe that before the 260s mark, the signal is adopting a certain behavior. Aside from the fluctuations observed throughout all the signal, we can interpret that in this first segment the mean of the signal is augmenting with time. This may suggest a linear trend that we are able to account for in our models. After the shock observed at the 260s we can see that the signal is regaining its previous pattern. The presented

p-values further supports the aforementioned interpretation. In fact, a p-value of 0.0402 in the KPSS supports the existence of a trend stationarity in the studied portion with a confidence level of 5%. Furthermore, we wanted to include this particular portion of the signal since it presents the unpredictable behavior illustrated through the chock. We will discuss in further sections a type of model that specializes in accounting for such behavior which is the Moving Average (MA) model.

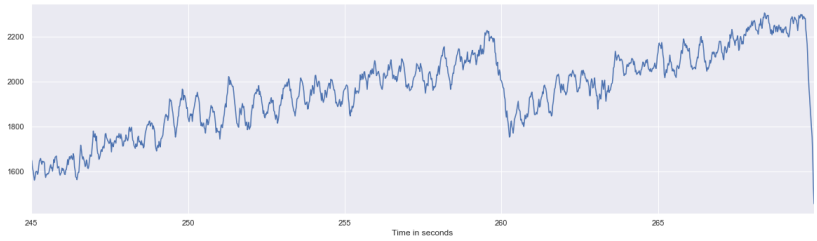


Figure 2.6: Evolution of BCG over a period of 25s with p-values of 1.668241113e-5 and 0.0402 for respectively the ADF and the KPSS tests.

2.4.3 Stationarity results in the BCG signal

The stationarity test has been conducted on the available dataset. This results were implemented using the statsmodels python package of time series analysis. The corresponding tests has been conducted on only the first subject. This result, however, can be generalized for the rest of the subjects. As mentioned before, the interpretation of the test results can be conducted through two main methods, mainly the p-values and through comparison with the critical values.

Table 2.2: Obtained results of the different conducted tests.

Test	p-value	test value	10%	5%	1%
ADF	1.57e-12	-8.067	-2.56	-2.861	-3.431
KPSS	0.1	1.932	0.347	0.463	0.739

Table 2.2 resumes the obtained results. Given the provided p-values we can safely adopt the non stationarity of the studied signal with a confidence level of 5%. If we want more detailed results as to determine the 1% confidence level, we can see that the obtained test values from the ADF test is less then the critical value at 1%. Which means that we are

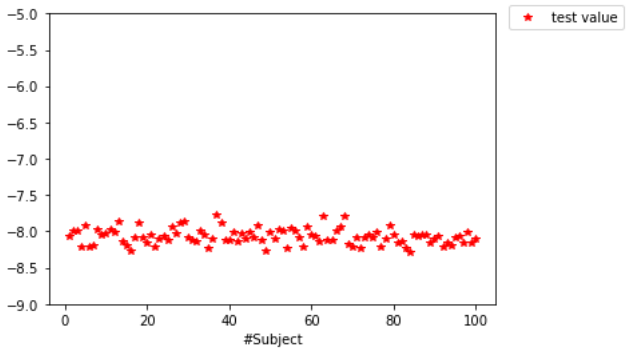


Figure 2.7: ADF values obtained from the different subjects(same critical values).

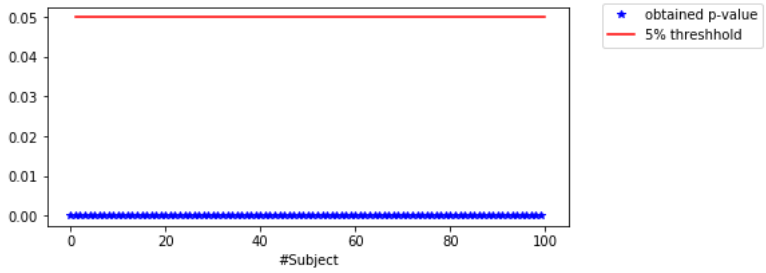


Figure 2.8: p-values of the ADF test obtained from the different subjects).

at least at the 1% confidence level when it comes to the ADF hypothesis. For the KBSS case, obtained p-value exceeds the 0.05 mark, which means that we can't reject the null stationary hypothesis. Test values that exceed the 1% critical value indicated that we are 1% sure of the aforementioned result.

2.4.4 Stationarity results in the complete dataset

In this section, we chose to generalize the stationarity results obtained from the subject at work on the complete available data. Figures 2.7 as well as 2.8 show the evolution of the p-values of the ADF test as well as the test values obtained across the different subjects. We can see that the p-values are below the 0.05 adopted threshold in the biomedical domain. Further interpretations on figure 2.7 with respect to the 10%, 5% and 1% critical values displayed in table 2.2 shows that this test is satisfied ti the extent of 1% significance. Figures 2.10 as well as 2.9 shows the obtained KPSS tests on the data set. Similarly to the specified use case in the previous section, these results confirm the stationarity hypothesis of KPSS.

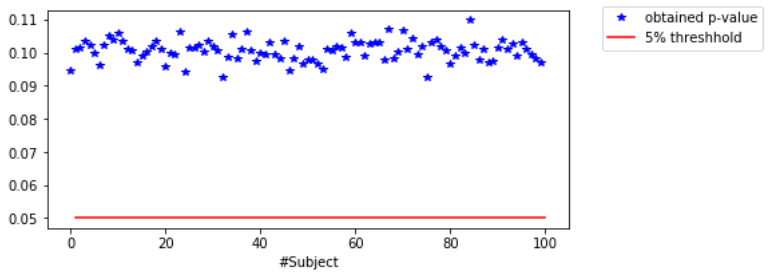


Figure 2.9: p-values of the KPSS test obtained from the different subjects).

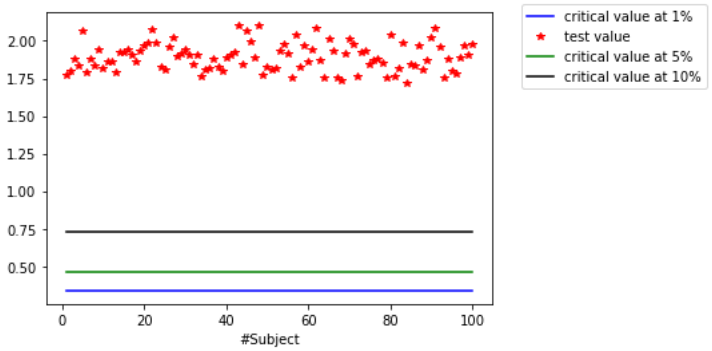


Figure 2.10: KPSS test obtained from the different subjects).

The results obtained in the trend example as well as the seasonality example were handpicked to showcase some non stationarity examples. However, results displayed in the table 2.2 shows that such examples are rare and carries no statistical significance. In the lights of these results, as well as the results displayed in figures 2.7, 2.8, 2.10, 2.9 we can safely adopt the ARMA model instead of the ARIMA model.

Chapter 3

BCG signal modeling

In this chapter we will discuss the different aspects that rules over the first part of our modeling: the forecasting. First, in subsection 3.1 we will present the theory behind the adopted ARMA model as well as the significance of each of its components, the it will be followed by results obtained using the adopted model in subsection 3.2 where performances of the model are adjusted as a function of its different hyperparameters. We chose to display the results in such s particular way as to highlight the variation of the performance as a function of different hyperparameters.
In subsection 3.4, we will provide a detailed study of the error metrics used in forecasting tasks as well as justify the choice of these metrics. We will present the validation scores with a certain novel approach of validation which is specific to the time series analysis entitled Walk-forward validation. Lastly, in section 3.4.3 we will present the final result of this first approach as well as some aspects to investigate in future works.

3.1 The ARIMA model

The Autoregressive Integrated Moving Average model (ARIMA) is a linear model that contains the two common principal components. the AutoRegressive (AR) component and the Moving Average (MA) one. Aside from the components that defines this model, the Integrated (I) term indicates the possibility to differentiate the model in order to surpass a certain type of non-stationarity. The general expression of an ARIMA model is

the following:

$$ARIMA(p,q,d) : (1 - \sum_{i=1}^{p} \rho_i L^i)(1-L)^d X_t = (1 + \sum_{i=1}^{q} \theta_i L^i)\epsilon_t. \tag{3.1}$$

Where p, q, d are the hyperparameters of the model, L is the back-shift operator, X_t is the signal at time t, ϵ_t is the error captured at the last iteration and $\{\rho_i, i < p\}$, $\{\theta_i, i < q\}$ are respectively the AR and MA parameters. The integration component of this equation introduced by the term $(1-L)^d$ is specifically added to the ARIMA models to take into consideration the seasonal behaviour of the time series. In fact, for $d = 1$, the modeled time series will be the return $X_t - X_{t-1}$instead of the original values of the BCG. This is the equivalent to modeling the gradient of the time series instead of the actual values. The solution that the integration proposes is the differentiation of higher order d in order to surpass the seasonality problems that can occur due to the respiratory fluctuations. Such behaviour was illustrated in figure 2.5. To further develop this point, we can utilize the Respiratory Rate (RR) in order to gain more insights over the choice of d. This can be subject to further works in the future. In this work, we chose to implement the simple ARMA model. Defined as:

$$ARMA(p,q) : X_t = \epsilon_t + \sum_{i=1}^{p} \rho_i X_{t-i} + \sum_{i=1}^{q} \theta_i \epsilon_{t-i}. \tag{3.2}$$

Where ϵ_t is white noise.

The AR component in this model express the current variation of the time series as a linear function of its previous state. The hyperparameter p is chosen as to obey to different criteria developed in the following sections. The AR parameters of this model $\{\rho_i, i < p\}$ are computed to minimise a certain error function.

The MA component in this model is implemented to take into consideration the different shocks that can occur on the time series. In fact, as displayed in section 2.4, the MA component will record, during the training phase the big value of the shock traduced by the high past error of ϵ_{t-1} which is non other than the value of $X_t - X_{t-1}$. This value as well as local information of the circumstances of its occurrence will be taken into

consideration in the next timestamps.

3.2 Hyperparameter tuning

One subject of huge importance in the modelling procedure is to find the optimal hyperparameters. This step carries huge importance since it can cause a large scale of problems, from high bias (under fitting) to high variance (overfitting) to non-convergence of the optimization problem which can induce a chaotic behavior of the model. It is well known that obtaining the optimal hyperparameters can be achieved through brute force methods such as search grid [69], or probabilistic method such as Bayesian optimization [66] more advanced methods uses reinforcement learning to find the optimal hyperparameters [36]. However, in the case of the ARMA models the hyperparameters are found using mainly 2 approaches. The first one considers the indicators of the model and the second one examines the Autocorrelation function and the partial autocorrelation function.

3.2.1 Tuning with autocorrelation function

Autocorrelation function

The autocorrelation function is defined as:

$$R_{XX}(\tau) = corr(X(t), X(t-\tau)) = E(X(t)X^*(t-\tau). \tag{3.3}$$

Where *corr* is the covariance function defined as:

$$corr(X, Y) = E(XY^*) \tag{3.4}$$

In the case of stationary time series, the autocorrelation function indicates the existence of correlation between samples in the signal or not. We are encouraged to add the components that are significant (the python package plots the significance levels at 5%). What this means is that to peaks that exceeds the aforementioned significance level, corresponds a p-value of less than 5%. This means that the information held by these peaks and in

particular their added value to the autocorrelation function is less than 5% likely to be subject of noise. However, considering all the significant peaks for the AR component is not a recommended approach.

Let's consider an AR (1) process, defined as :

$$X_t = \rho_1 X_{t-1} + \epsilon_t. \tag{3.5}$$

If we plot the Autocorrelation function of this synthetic AR(1) model we will observe very meaningful peaks until a very high lag. This is due to the fact that certain information included in X_{t-2} are propagated in the X_{t-1} and will propagate to the X_t. If we were to model our synthetically created time series X as an AR model, and we want to find the best fitting hyperparameter using the autocorrelation function. The found results in the autocorrelation plot displays multiple peaks at very advanced lags. Thus the sole use of the Autocorrelation function is insufficient and leads us to include unnecessary lags. The propagation of information discussed above is formulated using the kth order correlation coefficient expressed as follows:

$$f^k = \frac{E(X_t X_{t+k})}{E(X_t X_t)} \tag{3.6}$$

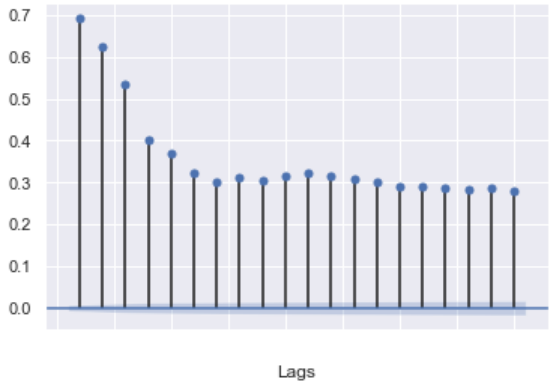


Figure 3.1: Autocorrelation function of the BCG signal

Figure 3.1 shows the variation of the autocorrelation function (Acf) of the studied signal. In blue, we find the significance level that obey to the 5% hypothesis discussed above. We can see that the Acf presents significant peaks until the 20 lag. This particular fact is intuitive and falls in line with the explanation we provided in the last paragraph

and thus we will only consider the first 5 lags in the MA component. At this point, we cannot derive a complete conclusion of the choice we made. This approach as well as the following only provides us with an intuition of the best number of lags to include. Conclusions concerning the goodness of fit will be provided while using the likelihood and residual analysis.

Partial Autocorrelation function

On the contrary of the Acf, the Partial Autocorrelation function (PAcf) contains peaks that reflects information about only the corresponding lag. This is done by including an orthogonal project operator in the definition of the PAct that eliminates the effect of the unwanted components. Thus the lags present in the PAcf can be accounted for in the process of the choice of the AR hyperparameters.

$$PAcf(\tau) = corr(X_t - P_{t,\tau}(X_t), X_{t+\tau} - P_{t,\tau}(X_{t+\tau})). \tag{3.7}$$

Where $P_{t,\tau}$ is an inner product defined in [18] as the orthogonal projection onto the linear subspace of Hilbert space spanned by $x_t ... x_{t-\tau}$

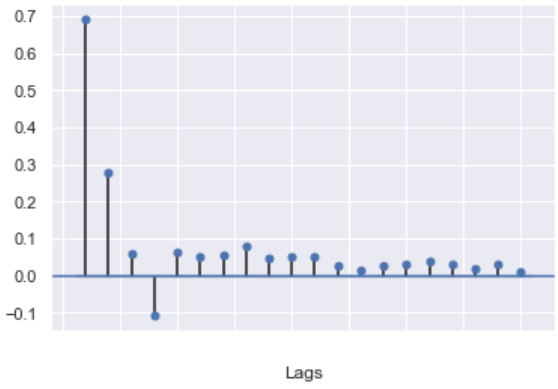


Figure 3.2: Partial autocorrelation function of the BCG signal

Figure 3.2 shows the evolution of the PAcf of the same BCG signal previously displayed. We can observe that, above the 5th component, the different peaks, whilst still statistically significant, are diminishing drastically. We will only include until the fifth component of the signal since with the AR process, the latter will be accounted for in the

other component.

3.2.2 Tuning with Likelyhood function

The main criteria used to investigate the goodness of fit of the model is the likelihood. We will use this criterion as well as the Akaike Information Criterion (AIC) [13] and Bayesian Information Criterion (BIC) [48] to compare between the different models. Fig-

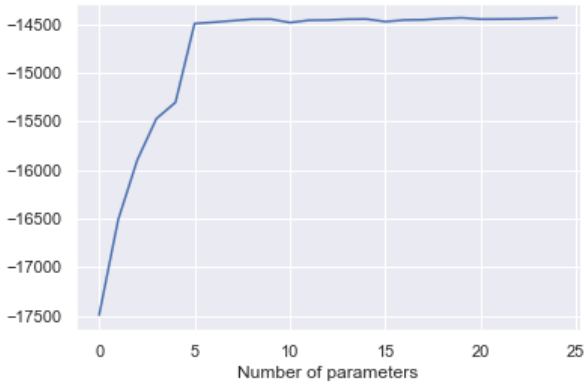


Figure 3.3: Evolution of the log likelihood over the number of parameters to fit

ure 3.3 shows the evolution likelihood as a function of the number of parameters of the model(MA+AR). This shows that after the fifth parameter which corresponds to the model ARMA (2,2) the log likelihood function is indifferent and showcases a quasi-constant behavior. If we were to consider only this criterion and the Akaike philosophy, we should consider the model ARMA (2,2). This philosophy states that given enough parameters, we can model every phenomenon. However, the trade-off comes in the overfitting aspect of things. Thus, we should consider the model that presents the best Likelihood and is the simplest since it will generalize better than overcomplicated models.

The AIC is another criterion based on the latter philosophy. It punishes the complexity of the model and thus gives us a good trade-off between goodness of fit and overfitting. The AIC is defined as follows:

$$AIC = 2k - log(L). \tag{3.8}$$

Where L is the likelihood of the model and k is the number of parameters.

The BIC is very similar to the AIC criterion in the sense that it punishes the high complexity of the model. The BIC is however more restrictive when it comes to the complexity of the model introducing a term depending on the scale of the training dataset.

$$BIC = klog(n) - 2log(L). \tag{3.9}$$

Where n is the number of training points.

In the contrary of the methods used when interpreting the log likelihood [62] to deduce the best fit of the model, we want to take the minimum of the AIC values.

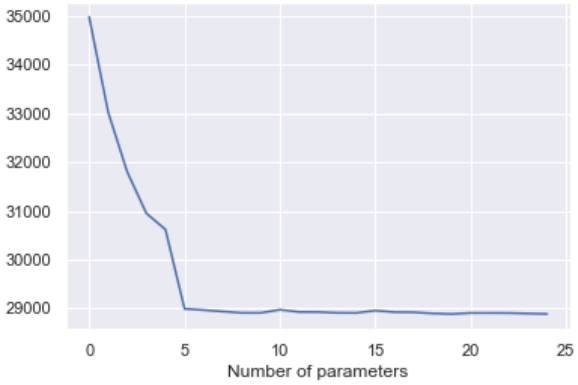


Figure 3.4: Evolution of the AIC over the number of parameters.

Figure 3.4 shows a similar behavior of that of the log likelihood. In fact, due to computation limitations and the rule of thumb in the statistics domain the number of parameters should not exceed 27 [35]. However, if we were to compute more number of parameters, we will observe the resurgence of the curve for very high values of parameters. The main issue resides in the fact that these criteria did not inform us much about the model selection. We can see that our model should include more than 5 components, we cannot however pinpoint the perfect model. In the next section we will present a rigorous method to select the model.

3.2.3 Residu analysis results

The residue is defined as

$$Res = X_t - \hat{X}_t. \tag{3.10}$$

Where X_t is the actual value of the time series at the step t and $\hat{X}_t$ is the forecasted value of the time series at the step t using the adopted ARMA model. The perfect fit of our model corresponds to a white noise residual. Thus we can observe the residuals, interpret their plots and deduce how close they are to white noise. A time series is White noise if its autocorrelation function presents no peaks at lag 0 and otherwise. We will utilize this property as well as the residue of the different models fitted above to extract the best model.

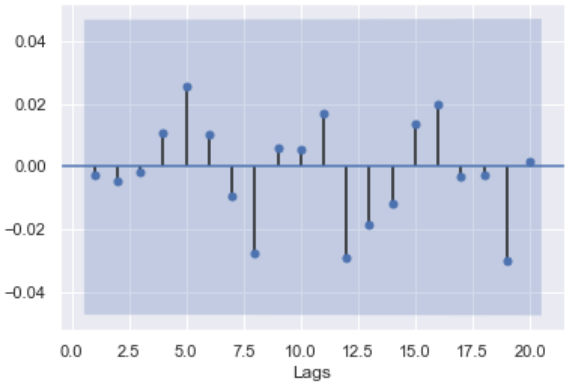


Figure 3.5: Acf of the Residual of the model.

After examining the different fitted models, we found that the most simplistic model that obeys to the white noise residual is the model ARMA (3,4) and thus we can say with confidence that our model is well fit since there's no significant component left in the residuals as shown in Figure 3.5. It shows that all the lags present in the residual are below the 5% significance level. This result further supports the choice of the corresponding hyperparameters.

Recent studies [50] have shown that convergence to the second-order stationarity in a non convex environment is an NP-hard problem. NP-hard stands for non deterministic polynomial time hardness. This means that there's no polynomial time verification for

these types of problems which differs from NP-complete problems that can be verified in polynomial time but not solved in such a time. If we want to be technical, the previous statement is not entirely true and not false either. In fact, one of the millennium problems concerns the P vs NP problem. Which means if someone can provide insights about this problem, in particular prove that problems that are verifiable in polynomial time are also solvable in polynomial time or the opposite has a very attractive price of 1million USD.

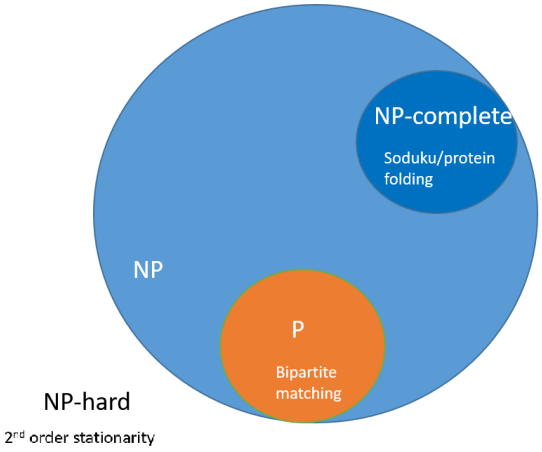


Figure 3.6: Stationarity and its complexity.

Figure 3.6 shows the partition of the aforementioned dilemma. As we can see the second order stationarity is an NP-hard problem. A funny part about this figure comes with the definition of reduction which means that finding a solution to a problem A can imply a solution to another problem B. That being said, every NP-complete problem can reduce an NP problem and thus finding a solution to Sudoku can reduce us to finding a solution to protein folding and thus curing cancer.

3.3 ARMA transformation using Box-Cox

One of the biggest critiques of the ARIMA model is its simplistic representation. In fact, the hypothesis of the ARIMA model is that the data can be modeled with a linear process, and the residue found will be a white Gaussian noise. However, this hypothesis is quite strong and rarely happens in real conditions. Thus statisticians are more often then not satisfied with only white noise residue which is the obtained result shown in figure 3.5.

The absence of peaks above the significance level defined as always at the p-value of 5% mark as of all medical application, shows that our residue is likely to be a white noise and thus somewhat satisfies the modelling conditions in the ARIMA definition.

After an exhaustive literature review, we found that ARMA models, granted they only model linear relationships, can be upgraded to include power relationships using mathematical transformations. These transformations are used in such cases to upgrade the model to be able to represent more complicated relationships, since in the reality, only rare cases are represented by linear dependencies. One such transformation is the Box-Cox transformation [11] [65]. The Box-Cox transformation is defined as follows:

$$X_t = \begin{cases} log(X_t) & \text{if } \lambda = 0 \\ \frac{(X_t^{\lambda}-1)}{\lambda} & \text{else.} \end{cases} \tag{3.11}$$

Where X_t is the time series at question and λ is the Box-Cox parameter to determine. It is obvious to point that for this transformation to be valid, a prior preprocessing has to be done in order to make the term X_t strictly positive. This can be easily achieved by simply subtracting the minimal negative value from our time series. This will result in the vertical shift of our signal to the positive domain. This transformation, however, does not affect our predictions since forecasting is about predicting the pattern in the signals not the range of the values itself. This value is chosen in a way to maximize the likelihood function while producing a statistically significant results (p-value inferior to 5%).

The results will be analyzed in a different way in this section. In fact, we want our Box-Cox transformation to ensure a better fit of our model, by leaving a Gaussian residue. What we propose is the plot of the Quantile-Quantile (QQ plot) of the residue of our fitted model before and after the transformation.

The QQ plot plots the different quantiles of the obtained distribution (in this case the residue) against the quantiles of the normal distribution. If the resulted plot is close to the first bisector, then we can safely say that we are at the presence of the normal distribution.

In figure 3.7, we can observe that our plot drifts from the first bisector plot (in

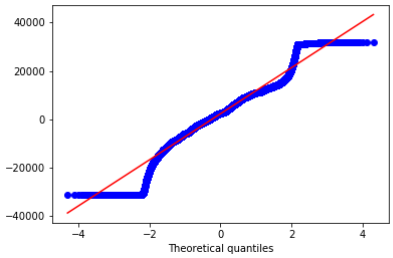


Figure 3.7: QQ plot of residue *without* Cox-Box.

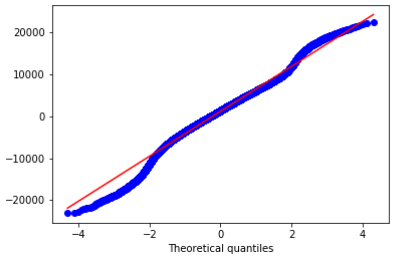


Figure 3.8: QQ plot of residue *with* Cox-Box.

red) this is due to the fact that the residue obtained, is a white noise in the statistical sense but fails to check for the normality property.

However, in figure 3.8, we can observe that the plot of the transformed residue is very close to the first bisector which means that we can safely say that the fitted model leaves a Gaussian white noise in the residue, which is the strong hypothesis aforementioned.

3.4 Performance and evaluation

In this section, we will present the results and the performance of our model. First we will define the different error metrics used in this study and then present the method the we employed to present the results. This latter is named in the literature “the walk forward validation”.

3.4.1 Error metrics

In this section we will explore different error metrics used in this work. This section is particularly important in our work since the performance will be measured using these metrics.

This subject is very interesting since following an exhaustive literature review, we can safely conclude that depending on the error metric, the application and the conclusions can vastly differ and vary. Some of the adopted metrics are defined as follows:

$$MSE = \frac{1}{N}\sum_{i=1}^{N}(X_i - \hat{X}_i)^2. \tag{3.12}$$

$$RMSE = \sqrt{(MSE)}. \tag{3.13}$$

$$MAE = \frac{1}{N}\sum_{i=1}^{n}|X_i - \hat{X}_i|. \tag{3.14}$$

Where N is the frame length, X_i is the actual value of the signal at step i and $\hat{X}_i$ is the forecasted value at step i. Some of the metrics that we are going to talk about are quite familiar such as the Root Mean Squared Error (RMSE) or the Mean Absolute Error (MAE), these particular metrics are unscaled since their values can theoretically reach very high proportions and they do not carry a particular significance as to how far are we from the actual values. Some more advanced errors has been developed to abstain from this particular issue of the RMSE and MAE such as the Mean Absolute Percentage Error (MAPE) and the symmetric Mean Absolute Percentage Error (sMAPE)[16][6]. A formal mathematical definition will be provided to such metrics as well as the choice done for this work and critics. RMSE provides a value that's comparable to the values taken by the BCG signal. A very similar error is the MAE. Both the MAE and the RMSE can be used to diagnose the model, while the RMSE computes the quadratic differences the MAE computes the linear difference so the difference between these two can provide an insight on the variance of the errors in the samples. The bigger the difference between these two, the bigger the variance in the recorded errors.

The issue with these metrics is that the values of these errors are not scaled, so we cannot cultivate a consistent information from its values and thus the selection methods based on these errors are heuristic and most of the times inconsistent when taken outside of its environment[17], thus the need for scaled errors with values that provides physical significance.

The most employed scaled error is the Mean Absolute Percentage Error (MAPE)

defined as:

$$MAPE = 100 * \frac{1}{N} \sum_{i=1}^{N} |\frac{X_i - \hat{X}_i}{X_i}|. \tag{3.15}$$

This error is scaled in the sense that its values represents a percentage of deviation of our forecast from the actual values. This is the most commonly used error metric in forecasting problems. However, the MAPE presents several drawbacks [46]:

1. Indefinite for null actual values X_i
2. It presents an asymmetry between X_i and $\hat{X}_i$.
3. It presents a second asymmetry in the case of positive values of the time series. The lower limit 0 cannot exceed an error of 100% while there is no upper limit when it comes to over forecasting.
4. The metric presented is vulnerable to outliers. If our model reports a single error and that error is very deviated from the actual value, the error presented above will present the same effect as for a model that commits a lot of less significant mistakes. In our case, we want our model to favor the first measure because we can handle outliers better then spread-out errors and thus the need for a more tolerant metric towards these outliers.

The introduction of the sMAPE was as an answer to the listed points above. Defined as:

$$sMAPE = 100 * \frac{1}{2 * N} \sum_{i=1}^{N} \frac{|X_i - \hat{X}_i|}{|X_i| + |\hat{X}_i|}. \tag{3.16}$$

The sMAPE resolves the asymmetry issue raised in the 3rd point above as well as the outlier's lenience with the introduction of the forecasted value in the denominator. However, this metric still presents a bias when it comes to under forecasting as opposing to over forecasting. We will however employ this particular metric in our results.

3.4.2 Walk-forward validation

In this section, we will discuss a very important validation process. The walk forward validation [19] is a sophisticated validation approach for time series.

When it comes to time series, it is a very hard and complicated process to evaluate, at least compared to cross sectional models. While the latter has very sophisticated models at its own such as k-fold, these methods seems to be lacking when it comes to panel models. One key aspect of these models is the time dependency. It is judged very important for panel models in particular time series models to take into consideration a very key aspect of these models, an aspect that defines them which is the time. While cross validating such models, the order of the samples as well as its the general environment needs to be provided to nest a fair environment to validate the model. In this regard, we are talking about the order of the samples taken into consideration, but a little bit more then that. In fact, it is shown in [19] and in [14], that panel models can be easily mistaken if not provided exactly the same environment in the validation process. A key example is found in Bradycardia. One aspect of detection of such abnormality is the succession of the frames that contain very low HBR. If we, for example take the k-fold cross validation to evaluate such process sample-by-sample, after shuffling the data we will lose the information about the low HBR and the characteristic of such sample. This is a very dangerous mistake since in this case we will not be able to judge the performance of our model.

Even if we take the validation process frame-by-frame and given that we can not be provided with the perfect frame length especially for constant length throughout the project, we can always find ourselves in the same case of the previously mentioned problem.

What the walk forward validation process proposes is the continuous validation of our model. This method consists of presenting an initial window of training. In the common cases the length of this window is the minimum amount of data points that can train the model, each iteration we retrain the model by predicting the values of the next p data points and computing its errors followed by adding them to the training set and retraining the model. This process is repeated until we exhaust the available dataset, we judge the obtained accuracy as satisfactory or we use more data than required and thus our model starts to deteriorates (this is the case of ARIMA models that presents an average performance with huge datasets). As mentioned before in section 3.4.1, the

adopted error we will use is the previously defined sMAPE since it provides the sought advantages.

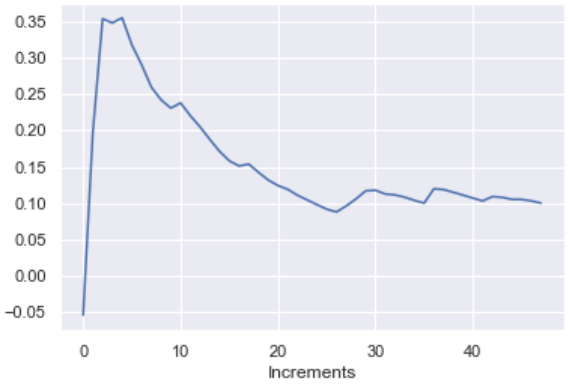


Figure 3.9: Evolution of the sMAPE of the transformed ARMA model over the training size.

Figure 3.9 shows the evolution of sMAPE over the different increments. The value of p chosen here is 50 to be conform with the physical reality of one second increment (the sampling rate in our data is 50Hz). It is worth mentioning that the initial training dataset employed in this study is different from the minimum required data points to train our model ARMA (3,4). To be precise our model requires only 7 data points to train but we omitted the corresponding initial part to show the significant fluctuations of the sMAPE. The first increments present very high errors (35%), these values decreases after and stabilizes around the 10% mark.

3.4.3 Predictions and discussion

In this section we will provide the plot of the prediction of the resulting model as well as present some limitations and improvement paths.

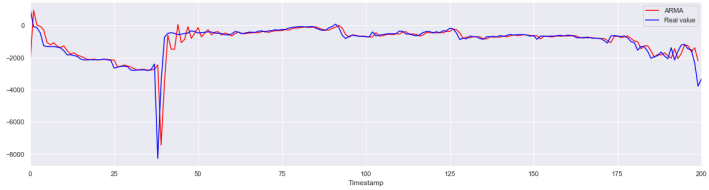


Figure 3.10: Prediction of 2 seconds ahead for the obtained model.

We can see that our model presents decent forecasts. The remarkable variations are witnessed around the 40 Timestamp mark.

After accounting for the occurring chock, our model showed certain fluctuations over the actual value (in blue). While these fluctuations are somewhat negligible, this behavior can present certain issues and problems especially when it comes to the medical application. In fact, such predicted behavior can be interpreted as tachycardia or tachypnea which are very common phenomenon in the medical field. We can see that our model regains its stability around 60 Timestamp mark which can be the silver lining of this study.

In fact, the fluctuating portion that corresponds 20 Timestamps corresponds to less than half a second and thus will not have a devastating effect on the performance of our model since we can always employ a median filter to the result or because of the set up itself. In fact, we can, as mentioned before, set a certain confidence level in our obtained results knowing that the overall performance of our model will be satisfactory. If we want to provide a general critique to our model; we wanted to utilize a linear model for our BCG forecasting step since little to none works has been conducted in the field of forecasting. This choice is the fruit of out intuition since the scientific way of doing things is to not jump to conclusions with the hyper complicated and sophisticated models but to start off with simple models instead. However, if we want to surpass the 7% gap of sMAPE shown in figure 1.6, we have to deploy more complicated models such as LSTM models. These however are very demanding when it comes to hyperparameters tuning and similarly to the ARMA models a single misstep can result in non satisfactory performance.

In this chapter we performed the statistical modeling of our forecasting part. A variant of the ARIMA model has been adopted which is the ARMA model. The performance of this model as well as the hyperparameter tuning was made using different methods which is a very attractive property of statistical modeling. First, we chose the lags likely to carry significant input using the autocorrelation and the partial autocorrelation. The latter only provides an intuition concerning the lags to include. Further studies done implemented the performance of different models as a function of the number of parameters and the autocorrelation of the residue was used to confirm the white noise

hypothesis. However, the biggest limitations of the ARMA models resides in their linearity. The impact of such limitation was observed through the QQ plot of the residue, and taken care of using Boxcox transformations. We will discuss further more the limitations of the aforementioned model. The forecasted result obtained, will be fed into the event detection LSTM-based model to detect the abnormalities within the frame.

Chapter 4

Event detection

In this section, we are going to work on the detection of the cardiac abnormality of the patient. As mentioned before, our work consists of basically two parts; the first involves the forecasting of the BCG signal done in chapter 3, the second part of our work consists on the detection of abnormalities within the forecasted time series. This will help us predict the cardiac abnormalities that exists in the future. In this section, we will define the proposed solution based on deep learning (DL), provide a mathematical formulation of the hyperparameters tuning of our problem and provide classification results. A particular emphasis has been put in the search of the optimal hyperparameters since the main challenge of DL resides in the choice of these hyperparameters. The results will be provided in the last section of this chapter.

4.1 Definition

The adopted architecture is a Long Short Term Memory (LSTM) based architecture with Fully Connected (FC) layers. We will try to provide an explanation on how the LSTM cell works.

In the figure 4.1, c_t is the cell state, h_t is the hidden state, x_t is the input. Tanh is the hyperbolic tangent function, it squished values between [-1,1] and σ is the sigmoid

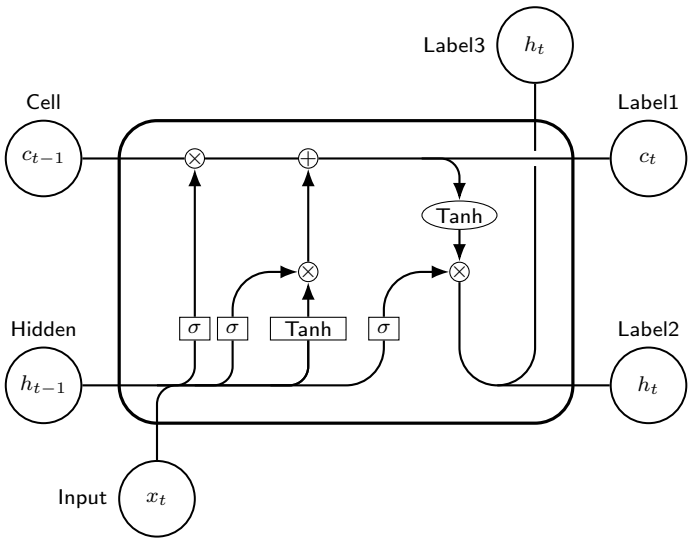


Figure 4.1: Basic LSTM cell

function defined as:

$$\sigma \colon R \to [0,1]$$
$$x \mapsto \frac{1}{1+\exp -x}.$$

The added value of the LSTM resides in the definition of the different states.

The core concept of LSTM's are the cell state, and it's various gates. The cell state act as a transport highway that transfers relative information all the way down the sequence chain. It can be considered as the "memory" of the network. The cell state, in theory, can carry relevant information throughout the processing of the sequence. So even information from the earlier time steps can make its way to later time steps, reducing the effects of short-term memory. Information gets added or removed to the cell state via gates. The gates are different neural networks that decide which information is allowed on the cell state. The gates can learn what information is relevant to keep or forget during training.

The forget gate:

Decides what information should be thrown away or kept. Information from the previous hidden state and information from the current input is passed through the sigmoid function. Values come out between 0 and 1. The closer to 0 means to forget, and the closer

to 1 means to keep.

The input gate:

At this level, the new input x_t as well as the previous hidden state h_{t-1} into a sigmoid function. The latter will decide which is important and relevant to the current task and score it between 0 and 1. After deciding the relevant information for this timestamp, the current input is kept to be included in the next timestamps with the sigmoid function. A sigmoid function is however utilized to refrain from the exploding gradient problem.

The cell gate:

We can look at it this way, the cell state concerns the main output of out cell at this timestamp. This output is denoted $Label1$ in our figure is the classical outcome of an LSTM. This is calculated as mentioned in the previous gates, by the use of the hidden state, the input state and the last cell state.

The output gate:

Decides what the next hidden state should be. The hidden state is also used for predictions. First, we pass the previous hidden state and the current input into a sigmoid function. Then we pass the newly modified cell state to the tanh function. We multiply the tanh output with the sigmoid output to decide what information the hidden state should carry. The output is the hidden state. The new cell state and the new hidden is then carried over to the next time step.

There's however a lot of variants of this architecture [40]. These variants concerns mainly the aforementioned architectures. Some works outputs both the hidden state and the cell state at each timestamp [71]. Some other works gets rid of the cell state and outputs only the hidden state. Transforming the classic 3-gate architecture to a 2-gate architecture with only an update gate and a reset gate. This is called a Gated Recurrent Unit (GRU) [68].

The LSTM cell can handle decent far look backs into the past and this fact is very seducing especially when it comes to our application. What we implemented in this work is a hybrid variation of LSTM cells and FC cells. This is done with respect to the works done with CNN architectures. We expect the LSTM cells to perform the feature

extraction phase automatically and this extract the best features to feed to the FC layers. The architecture of our solution will be provided in the next section.

4.2 Proposed solution

4.2.1 Model architecture

In this section we will talk about the proposed solution to the event detection problem. The adopted architecture is called many-to-one. A 200 sample-length frame is fed to the first LSTM layer that contains 200 cells. After propagating the architecture of the proposed solution (4 LSTM cells and a FC network) only a single output will be extracted from the frame which is the existence/not of the heart abnormality.

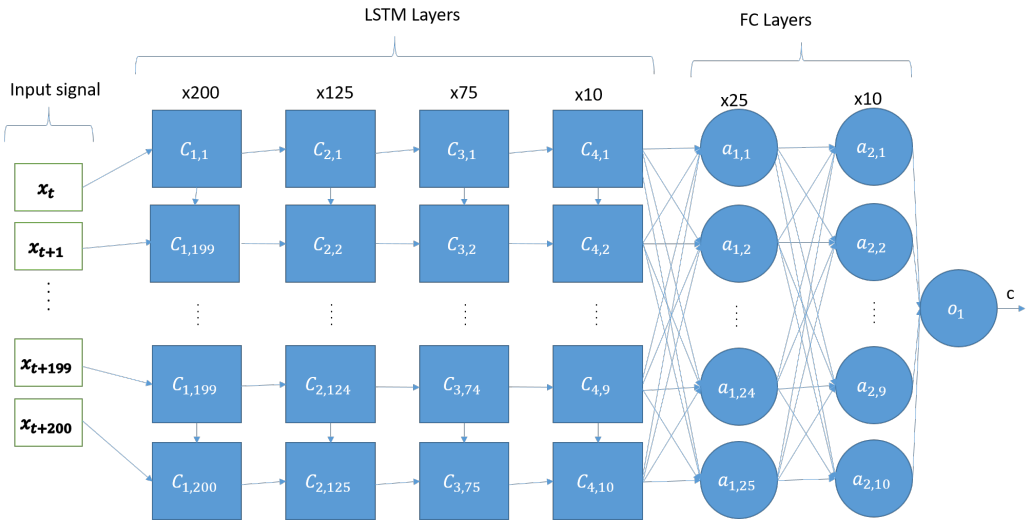


Figure 4.2: Adopted hybrid architecture.

The aforementioned architecture is described in figure 4.2. Our DL model was trained with 4 LSTM layers and 2 FC layers. The LSTM layers will play the role of the automatic feature extractors [63]. Each of the outputs of the first 3 LSTM layers is fed forward using the Keras option $returnsequence = True$. The result is fed to a two-layer FC network. The output of the signal is state value that provides the probability of the abnormality. This output is calculated using a modified Binary cross-entropy loss function in the output layer o_1. More details about the loss adopted as well as the activation function will be discussed in the next section. The architecture provided in

figure 4.2 is not however complete since we added dropout layers after each LSTM cell. We chose to not include this detail for the sake of the simplicity of the architecture. The latter was manually tuned due to lack of available time, however, different approaches can be used to chose the architecture of the model. One of which is the Bayesian optimisation [41].

4.2.2 The magic within

Following the automatic feature extraction process explained in the previous section, a feature space $F = \{F_1, ..F_{10}\}$ will be fed into the FC layers. In other terms, from input signal $x_t..x_{t+200}$ the LSTM layers will extract 10 temporal features which will represent the input of our FC layers. The generation of this feature space is computed using the Back Propagation Through Time algorithm (BPTT) which is the mainly adopted algorithm for RNN application [67].The neural network represented by the FC layers will compute its weights using the classic Feed Forward (FF) and Back Propagation (BP) algorithm.
A detail worth mentioning in the loss function and the activation functions used in this work. The loss function adopted to compute the weights of the FC layers is a weighted Binary cross-entropy. In fact, the data given has an unbalance proportion of 43000 to 4300 after the data augmentation (see section 2.2). The formal definition of this function is:

$$BCE_w = \sum_{i=1}^{2} w_i y_i log(\sigma(s_i)). \tag{4.1}$$

Where y_i is the label of a given sample, w_i is the weight of the class, s_i is the output of the network of the given sample and σ is the activation function adopted in the output layer. The weights of the classes is the second method adopted to overcome the skewness problem of our dataset. In fact, these weights are employed in the calculation of the loss function to put more emphasis on the underrepresented class. Generally, these weights are taken in consideration to the ratio of the dataset. However, the choice of the exact weights has to take into consideration tensor calculation optimisations within the backend employed by Keras: Tensorflow2.0 [32]. In other words, very low or high values of the weights can mess up with the tensors in the BP algorithm. Keras provides a method to

automatically calculate the weights of the classes using the $class_weights$ function. The output weights obtained in our example are: $w_0 = 5.62$ and $w_1 = 0.57$. It is worth mentioning at this point the the class denoted 0 is the negative class that denotes the existence of an abnormality.

The activation function adopted in the output layer previously noted σ is the sigmoid function above. This function is the employed activation in the output layer and provides us with the probability of cardiac abnormality. The other activation function employed throughout the architecture is the rectified linear function which is proved to provide the best results in deep learning architectures [26].

4.3 Results

In this section, we will provide the classification results obtained. We want to present the obtained results in a way to display the added value of the low-level data augmentation deployed in this work as well as the effect of the model weights. We will however start by defining the classification metrics

4.3.1 Classification metrics

The classification results can be extracted from the confusion matrix. Other the well know metrics we define a weighted accuracy that takes into consideration the cardinal of each class. This particular metric is efficient when it comes to classifying unbalanced data sets[24] [27]. The obtained results will be presented in table 4.1.

- $Accuracy = \frac{TruePositives}{|D|} + \frac{TrueNegatives}{|D|}$
- $Weightedaccuracy = w_1 \frac{TruePositives}{|D|} + w_0 \frac{TrueNegatives}{|D|}$
- $Precision = \frac{TruePositives}{Truepositives+FalsePositives}$
- $Recall = \frac{TruePositives}{Truepositives+FalseNegatives}$
- $f1 = 2 * \frac{Precision*Recall}{Precision+Recall}$

Table 4.1: Obtained metrics of weighted and augmented model and baseline model.

	Baseline model	Data-sensitive model
Accuracy	99.7%	95.5%
Weighted accuracy	60.8%	99.5%
Precision	94.7%	99.8%
Recall	100%	95.5%
Criterion	0%	95%
f1 score	97.2%	95%

Table 4.2: Normalized confusion matrix of the obtained model.

Actual classes	No abnormality predicted	Abnormality predicted
No abnormality	96%	4%
Abnormality	5%	95%

- $Criterion = \frac{TrueNegatives}{TrueNegatives+FalsePositives}$

. Where True positives is the number of normal frames classified as normal, true negatives is the number of abnormal frames classified as abnormal and —D— is the cardinal of the available dataset. After observing the provided results in table 4.1, we can see that the accuracy of the baseline model is in fact higher that the accuracy of the improved model. This particular fact is the reason behind the implementation of the weighted accuracy. In fact, the baseline model is highly biased towards the positive class (inlier) and predicts its value each time. This can be observed through the recall value of 1 and Criteria of 0. The developped model overcomes the skewness bias in the data set and provides decent results 99.5% accuracy. The previously defined criterion takes into consideration the medical relevance of out setup. In fact, we judge it important that our model does not leave any abnormality unnoticed. Which means that we want the proportion of True Negatives (abnormality recorded) with respect to the total actual abnormalities (whether it was recorded or not) to be high. In this work, we judge a detection rate of 95% to be satisfactory. Further inspections of the model can be done using the confusion matrix provided in table 4.2. In fact, these results were obtained on the out of samples dataset also referred to as the validation data. After shuffling the data, a validation split percentage of 20% was chosen to evaluate the model. The training was done on 80% of the data and the test was done on the unseen 20%. This approach however is not the adopted approach in the medical fields. Leave One Out Cross Validation (LOOCV) [33] is

the golden standard in medical data science. However, due to the dataset at hand and its properties, especially its skewness, we judged it important to make use of every available abnormalities from the different subjects and thus this method is not applicable in our case since the training was done on all the subjects available to us.

The percentages observed in table 4.2 shows very promising results. Our model can detect the abnormalities with a high accuracy of 95%. As mentioned in the previous section with the definition of the *Criteria*, our model pays particular attention to the detection of abnormalities when they occur, leaving out only 5% of the cases. This is particularly important since it reflects the physical significance of the application. We judge it important that our model detects the different abnormalities when they occur so that the final solution will be able to avert disastrous consequences.

Chapter 5

Summary and Reflections

In this work we presented a solution to detect the different heart abnormalities of the patient. The aim of this work is to establish a framework for the detection of specific abnormalities in the future as well as the monitoring of the health condition. The proposed solution predicts at two-steps the existence of heart abnormalities one second ahead.

The first step of this work is data preprocessing. Three main characteristics of the available data set were studied, namely: the noisiness, the skewness and the stationarity. As per the noisiness of the signal, we proposed an unsupervised learning solution developed during the sophomore internship [9]. This solution in particular extracts two features from the signal and classify the latter as noisy or normal based on its periodicity.

The skewness of the dataset was tackled on two levels, first by performing two data augmentation techniques which is the low level solution. A higher level solution was utilized later on by implementing class weights in the loss function. This helped take into consideration the proportions of the available data in the training process and thus according more importance to the underrepresented classes.

A formal study has been conducted on the stationarity of the studied signal. This has been done with the use of two different tests, the ADF test and the KPSS test. We provided some examples of the non stationarity of our signal, mainly presenting cases of seasonality and trend. These two examples however had little to no effect on the stationarity of the provided signal since they carry no statistical significance. The latter result was provided through the different values of the two tests on all the subjects' BCG

signals. This particular point can be the subject of future studies and works.

The modeling part of our solution has been done on two levels.
The first level concerns the forecasting of the BCG signal. In this part we are provided with a signal frame and we are asked to forecast its values a second ahead in time. The choice of the adopted statistical model was done after a careful review of [45]; one of the leading references when it comes to forecasting. The choice of the ARMA model was done due to the novel aspect of this work and with respect to the easy interpretability of this latter model. Mathematical transformation was performed on the dataset in order to obtain normality. Different error metrics were studied and the performance of our model was provided using a specialized validation method for time series. Although the obtained model showed state of the art results, we are confident that better results can be achieved using more specialized models. In fact, the nature of the presented BCG signal shows heteroscedasticity. This technical term means that the variance of the signal is not constant throughout time. This is proved to disrupt the performance of the ARIMA model. Specialized models such as Generalized AutoRegressive Heteroscedasticity (GARCH) model takes into account this variable volatility of the signal by modelling the variance of the signal using an AR model.

The second level concerns the event detection. On this level, we are provided with the output of the first model which is the forecasted frame of 1s ahead and we are asked to predict the existence/absence of a heart abnormality. State of the art results were obtained using and LSTM and FC layers architecture. A particular criteria was defined to take into account the physical reality and emphasise on the importance of the detection of the different abnormalities when they occur. More developed mechanisms can be adopted in this part in order to refine the classification results, such mechanisms are the use of CNN layers as feature extractors, pretrained models and especially attention mechanisms [38]. The latter mechanisms is used to motivate the model as to put more emphasis on particular parts of the signal.

In the future work, different approaches can be adopted. One of which is to develop the current work as to include the aforementioned improvements. Other ways

to push forward this work is to include different inputs instead of only the row BCG signal. Such works can implement multivariate ARIMA models, ARIMAX or with the respiratory rhythm information the SARIMAX model. And finally given more available data, more advanced validation process can be conducted as to simulate a more realistic environment.

Bibliography

[1] Bradycardia — slow heart rate. american heart association. http://www.heart.org/heartorg/conditions/arrhythmia/aboutarrhythmia/bradycardia-slow-heart-rateucm302016article.jsp. accessed dec. 28, 2016.

[2] Homoud mk, et al. sinus tachycardia: Evaluation and management. https://www.uptodate.com/contents/search. accessed march 3, 2020.

[3] Medical dictionary. the freedictionary.com [acessed 25 june 2020].

[4] Albadr, M., and Tiuna, S. Extreme learning machine: A review. *International Journal of Applied Engineering Research 12* (01 2017), 4610–4623.

[5] Aldmour, I. What is 5g?. key technologies, challenges and impacts.

[6] Armstrong, J., and Fildes, R. On the selection of error measures for comparisons among forecasting methods. *J. Scott Armstrong 14* (02 2005).

[7] Azhaginiyan, S. Denoising of bcg signal using multi resolution analysis. *2019 5th International Conference on Advanced Computing & Communication Systems (ICACCS)*.

[8] Ben nasr, M., and Ben Jebara, S. *Heart beat rate and respiratory rate extraction from ballistocardiogram signal*. PhD thesis, Higher school of telecommunication of tunis, 2019.

[9] Ben Nasr, M., Ben Jebara, S., Otis, S., Abdulrazak, B., and Mezghani. Noisy frame classification based spectral ballistocardiogram analysis. *18th International Conference, ICOST 2020, Hammamet, Tunisia, Proceedings*.

[10] Ben nasr, M., Ben Jebaraand S. Otis, S., Abdulrazak, B., and Mezghani, N. Respiratory activity classification based on ballistocardiogram analysis. *18th International Conference, ICOST 2020, Hammamet, Tunisia, Proceedings* (2020), 79–88.

[11] Bicego, M., and Baldo, S. Properties of the box-cox transformation for pattern classification. *Neurocomputing 218* (08 2016).

[12] Bm, M., and Sharma, C. Bcg artifact removal using improved independent component analysis approach. *Indonesian Journal of Electrical Engineering and Computer Science 5* (01 2017), 130.

[13] Brooks, D. Akaike information criterion statistics. *Technometrics 31* (03 2012), 270–271.

[14] Cao, L., and Tay, F. Support vector machine with adaptive parameters in financial time series forecasting. *Neural Networks, IEEE Transactions on 14* (12 2003), 1506 – 1518.

[15] Carrion-i Silvestre, J., and Sanso, A. A guide to the computation of stationarity tests. *Empirical Economics 31* (02 2006), 433–448.

[16] Chen, C., Twycross, J., and Garibaldi, J. A new accuracy measure based on bounded relative error for time series forecasting. *PLOS ONE 12* (03 2017), 1–23.

[17] Clements, M., and Hendry, D. On the limitations of comparing mean square forecast errors. *Journal of Forecasting 12* (12 1993), 617 – 637.

[18] Dégerine, S., and Lambert-Lacroix, S. Partial autocorrelation function of a nonstationary time series. *Journal of Multivariate Analysis 87* (10 2003), 46–59.

[19] Falessi, D., Narayana, L., Thai, J. F., and Turhan, B. Preserving order of data when validating defect prediction models, 2018.

[20] Fan, X., Zhao, Y., Wang, H., and Tsui, K. Forecasting one-day-forward wellness conditions for community-dwelling elderly with single lead short electrocardiogram signals. *BMC Medical Informatics and Decision Making 19* (12 2019).

[21] Gao, J., Song, X., Wen, Q., Wang, P., Sun, L., and Xu, H. Robusttad: Robust time series anomaly detection via decomposition and convolutional neural networks, 2020.

[22] Gao, W., Xu, Y., Li, S., Fu, Y., Zheng, D., and She, Y. Obstructive sleep apnea syndrome detection based on ballistocardiogram via machine learning approach. *Mathematical Biosciences and Engineering 16* (06 2019), 5672–5686.

[23] Giovangrandi, L., Inan, O., Wiard, R., Etemadi, M., and Kovacs, G. Ballistocardiography – a method worth revisiting. *Conference proceedings : ... Annual International Conference of the IEEE Engineering in Medicine and Biology Society. IEEE Engineering in Medicine and Biology Society. Conference 2011* (08 2011), 4279–82.

[24] Glodek, M., Tschechne, S., Layher, G., Schels, M., Brosch, T., Scherer, S., Kächele, M., Schmidt, M., Neumann, H., Palm, G., and Schwenker, F. Multiple classifier systems for the classification of audio-visual emotional states. pp. 359–368.

[25] Hadjem, M., and Naït-Abdesselam, F. An ecg t-wave anomalies detection using a lightweight classification model for wireless body sensors. pp. 278–283.

[26] Hara, K., Saito, D., and Shouno, H. Analysis of function of rectified linear unit used in deep learning. pp. 1–8.

[27] Hassan, M., Hossain, M., Bailey, J., Macintyre, G., Ho, J., and Ramamohanarao, K. A voting approach to identify a small number of highly predictive genes using multiple classifiers. *BMC bioinformatics 10 Suppl 1* (02 2009), S19.

[28] Hatcher, W., and Yu, W. A survey of deep learning: Platforms, applications and emerging research trends. *IEEE Access PP* (04 2018), 1–1.

[29] Hatcher, W., and Yu, W. A survey of deep learning: Platforms, applications and emerging research trends. *IEEE Access PP* (04 2018), 1–1.

[30] Hilgard J, Ezri MD, D. P. Significance of ventricular pauses of three seconds or more detected on twenty-four-hour holter recordings. am j cardiol 1985; 55:1005.

[31] Hochreiter, S. The vanishing gradient problem during learning recurrent neural nets and problem solutions. *International Journal of Uncertainty, Fuzziness and Knowledge-Based Systems 6* (04 1998), 107–116.

[32] Holtz, S., Rohwedder, T., and Schneider, R. The alternating linear scheme for tensor optimization in the tensor train format. *SIAM Journal on Scientific Computing* (01 2011).

[33] Homrighausen, D., and McDonald, D. Leave-one-out cross-validation is risk consistent for lasso. *Machine Learning 97* (06 2012).

[34] Huang, B., and Kinsner, W. Ecg frame classification using dynamic time warping. *Canadian Conference on Electrical and Computer Engineering 2* (02 2002), 1105 – 1110 vol.2.

[35] Jackson, D. Revisiting sample size and number of parameter estimates: Some support for the n:q hypothesis. *Structural Equation Modeling: A Multidisciplinary Journal 10* (01 2003).

[36] Jomaa, H. S., Grabocka, J., and Schmidt-Thieme, L. Hyp-rl : Hyperparameter optimization by reinforcement learning, 2019.

[37] Kachuee, M., Fazeli, S., and Sarrafzadeh, M. Ecg heartbeat classification: A deep transferable representation.

[38] Kann, K., Cotterell, R., and Schütze, H. Neural morphological analysis: Encoding-decoding canonical segments. 961–967.

[39] Kansal, S., Bansod, P., and Kumar, D. A. Multivariate autoregressive model for ecg signal forecasting. *International Journal of Multivariate Data Analysis 1* (01 2017), 124.

[40] Kent, D., and Salem, F. M. Performance of three slim variants of the long short-term memory (lstm) layer, 2019.

[41] Klein, A., Falkner, S., Bartels, S., Hennig, P., and Hutter, F. Fast bayesian hyperparameter optimization on large datasets. *Electronic Journal of Statistics 11* (01 2017), 4945–4968.

[42] Lee, B., Han, J., Baek, H., Shin, J., Park, K., and Yi, W.-J. Improved elimination of motion artifacts from a photoplethysmographic signal using a kalman smoother with simultaneous accelerometry. *Physiological measurement 31* (10 2010), 1585–603.

[43] Lu, H., Zhang, H., Lin, Z., and Huat, N. A novel deep learning based neural network for heartbeat detection in ballistocardiograph. *Conference proceedings: ... Annual International Conference of the IEEE Engineering in Medicine and Biology Society. IEEE Engineering in Medicine and Biology Society. Conference 2018* (07 2018), 2563–2566.

[44] Ma, Z., and Chu, T. The bcg signal feature extraction and recognition based on the cumulative residual entropy. 526–528.

[45] Makridakis S, Spiliotis E, A. V. *Statistical and Machine Learning forecasting methods: Concerns and ways forward. PLoS ONE 13(3): e0194889* (2018).

[46] Meade, N., and Armstrong, J. Long range forecasting: From crystal ball to computer (2nd edition). *The Journal of the Operational Research Society 37* (05 1986), 533.

[47] Mushtaq, R. Augmented dickey fuller test. *SSRN Electronic Journal* (08 2011).

[48] Neath, A., and Cavanaugh, J. The bayesian information criterion: Background, derivation, and applications. *Wiley Interdisciplinary Reviews: Computational Statistics 4* (03 2012).

[49] Nedoma, J., Fajkus, M., Martinek, R., Kepak, S., Cubik, J., Zabka, S., and Vašinek, V. Comparison of bcg, pcg and ecg signals in application of heart rate monitoring of the human body. 420–424.

[50] Nouiehed, M., Lee, J. D., and Razaviyayn, M. Convergence to second-order stationarity for constrained non-convex optimization, 2018.

[51] Nyamapfene, A. The impending 5g era and its likely impact on society. *Journal of Independent Studies and Research - Computing 14* (01 2016).

[52] Otis, S., Mezghani, N., and Abdulrazak, B. Comparative study of heart rate extraction methods for a novel intelligent mattress. 93–98.

[53] Qingsong Wen, Liang Sun, X. S. J. G. X. W., and Xu, H. Time series data augmentation for deep learning: A survey. *Machine Intelligence Technology, DAMO Academy, Alibaba Group 500 108th Ave, Bellevue 98004, WA, USA.*

[54] Rangasamy, M., Balasubramaniam, A., Krishnarajan, D., Raviteja, A., Kante, N., and Kumar, N. Role of telemedicine in health care system: A review.

[55] Rao, S., El Ali, A., and Cesar, P. Deepsleep: a ballistocardiographic deep learning approach for classifying sleep stages. 187–190.

[56] Salehinejad, H., Sankar, S., Barfett, J., Colak, E., and Valaee, S. Recent advances in recurrent neural networks.

[57] Sana S. Zadawale, S. B. Ecg signal based heart disease prediction system using dwt and svm. *International Journal of Advanced Research in Computer and Communication Engineering 6* (07 2017).

[58] Seyedtabaii, S., and Seyedtabaii, L. Kalman filter based adaptive reduction of motion artifact from photoplethysmographic signal. *World Acad. Sci. Eng. Technol. 37* (01 2008), 173–176.

[59] Stone, D. Application of median filtering to noisy data. *Canadian Journal of Chemistry 73* (02 2011), 1573–1581.

[60] Stramer, O. *Seasonality.* 09 2014.

[61] Sun, Z.-G., Lei, Y., Wang, J., Liu, Q., and Tan, Q.-Q. An ecg signal analysis and prediction method combined with vmd and neural network. 199–202.

[62] Tripathi, S., Bhardwaj, A., and E, P. Approaches to clustering in customer segmentation. *International Journal of Engineering and Technology 7* (07 2018), 802.

[63] Tzinis, E., and Potamianos, A. Segment-based speech emotion recognition using recurrent neural networks.

[64] Vargas, R., and Ruiz, L. Deep learning: Previous and present applications. *Journal of Awareness 2* (11 2017), 11–20.

[65] Velez, J., Correa, J., and Marmolejo-Ramos, F. A new approach to the box–cox transformation. *Frontiers in Applied Mathematics and Statistics 1* (10 2015).

[66] Wang, Z., and Freitas, N. Theoretical analysis of bayesian optimisation with unknown gaussian process hyper-parameters.

[67] Werbos, P. Backpropagation through time: what it does and how to do it. *Proceedings of the IEEE 78* (11 1990), 1550 – 1560.

[68] Wielgosz, M., Skoczen, A., and Mertik, M. Recurrent neural networks for anomaly detection in the post-mortem time series of lhc superconducting magnets.

[69] Xiao, T., Ren, D., Lei, S., Zhang, J., and Liu, X. Based on grid-search and pso parameter optimization for support vector machine. *Proceedings of the World*

Congress on Intelligent Control and Automation (WCICA) 2015 (03 2015), 1529–1533.

[70] XIYA, M. Covid-19 and the critical shortage in critical care. [online] global health now. *¡https://www.globalhealthnow.org/2020-05/covid-19-and-critical-shortage-critical-care¿ [Accessed 23 June 2020].* (05 2020).

[71] Zhang, D., Peng, Q., Lin, J., Wang, D., Liu, X., and Zhuang, J. Simulating reservoir operation using a recurrent neural network algorithm. *Water 11* (04 2019), 865.

[72] Zhu, Y., and Huang, C. An improved median filtering algorithm for image noise reduction. *Physics Procedia 25* (12 2012), 609–616.